

DISORDERS OF THE MALE GENITAL/REPRODUCTIVE SYSTEM

@2025

Machiko Alfred

LEARNING OBJECTIVES

- At the end of this lecture series, Students should be able to describe the common **congenital anomalies**, **inflammatory diseases** and **neoplasia** in the:-
 - Penis
 - Scrotum
 - Testes
 - Epididymis
 - Prostate gland

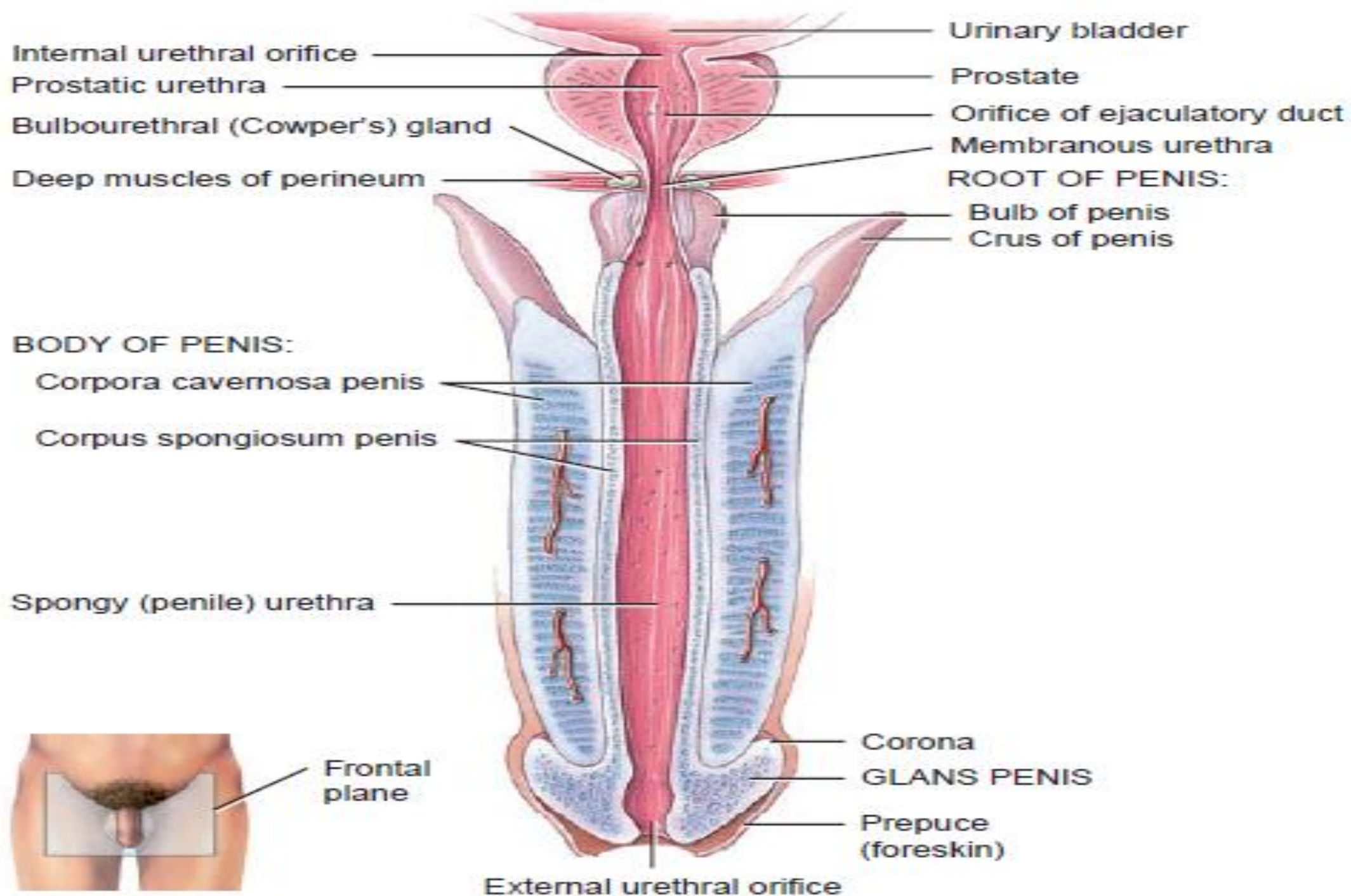
PENIS

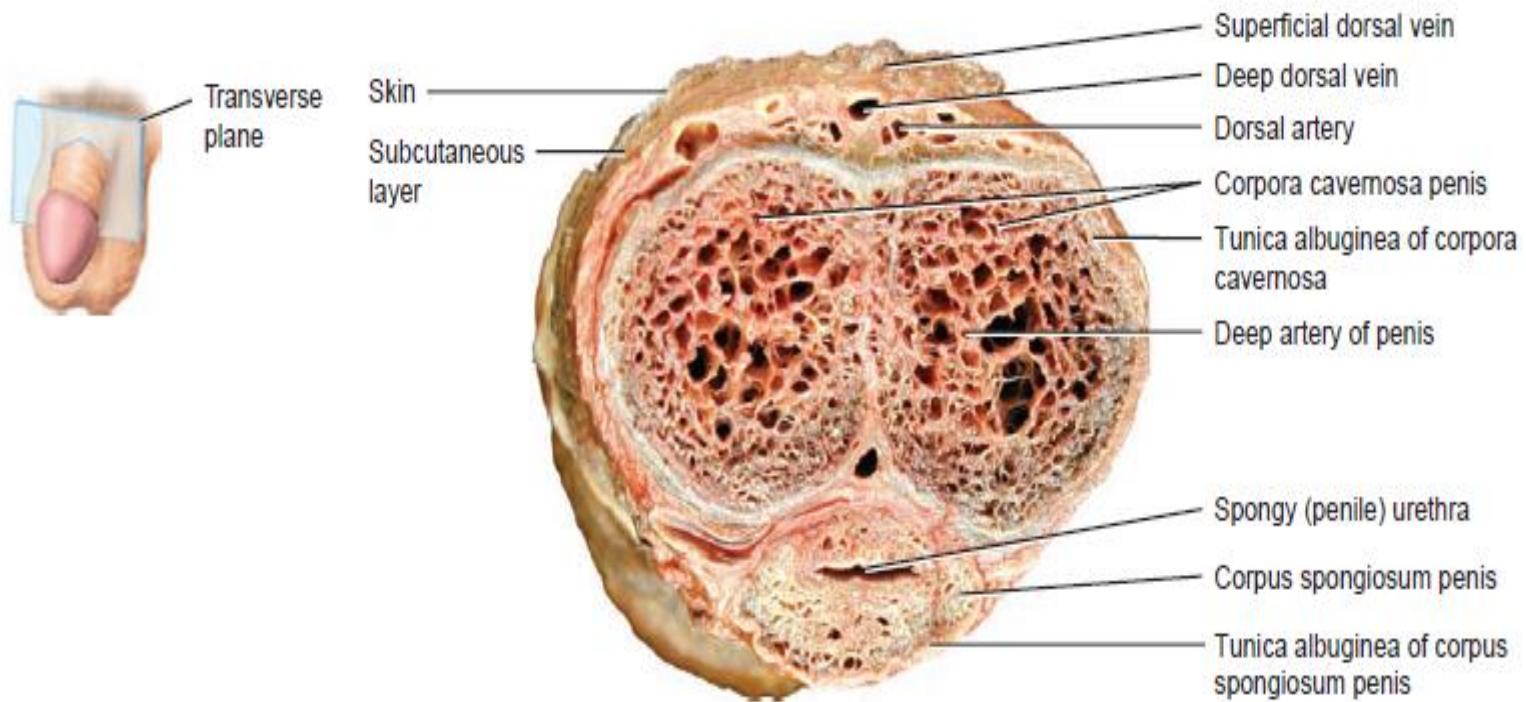
Normal structure

- The penis is covered by skin, foreskin (prepuce) and stratified squamous mucosa.
- The structure of penis consists of 3 masses of **erectile tissue**— two **corpora cavernosa**, one on each side dorsally, and the **corpus spongiosum** ventrally through which the urethra passes
- Expanded free end of the corpus spongiosum forms the **glans**
- The lumen of the **urethra** in sectioned surface of the penis appears as an irregular cleft in the middle of the corpus spongiosum

CONT'D

- In the prostatic part, it is lined by **transitional epithelium**, but elsewhere it is lined by **columnar epithelium** except near its orifice where **stratified squamous epithelium** lines it.

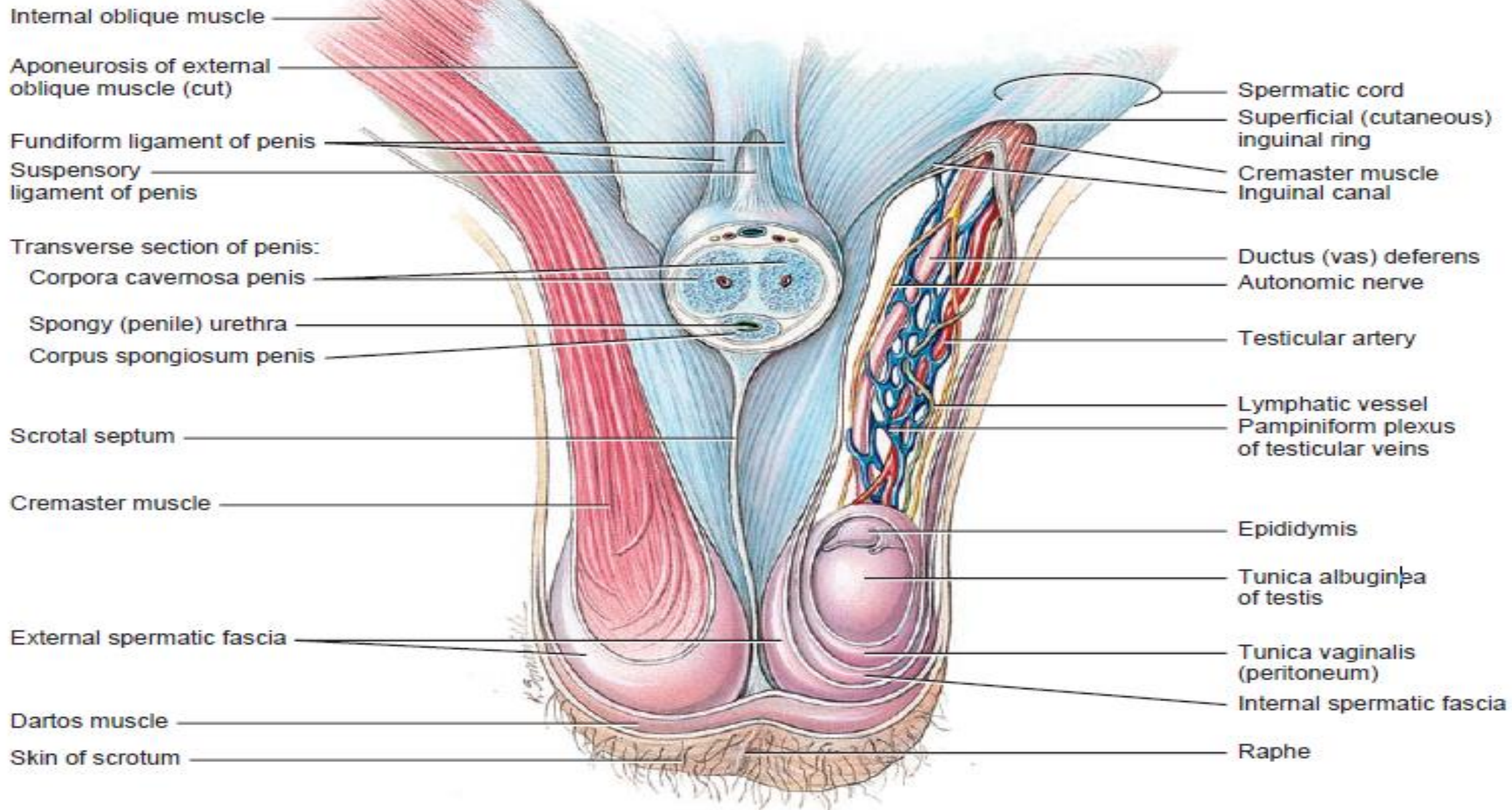


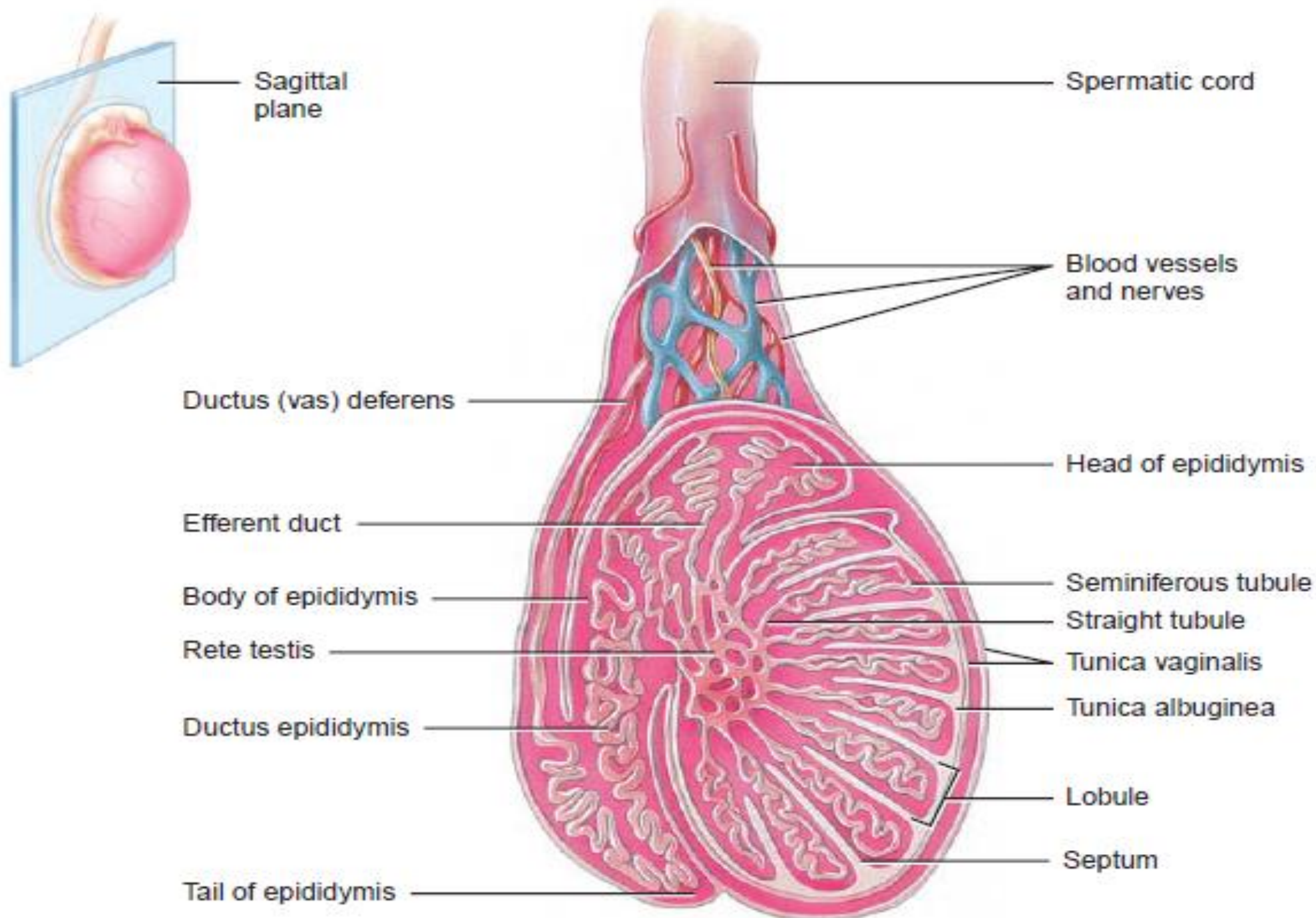


TESTIS AND EPIDIDYMIS

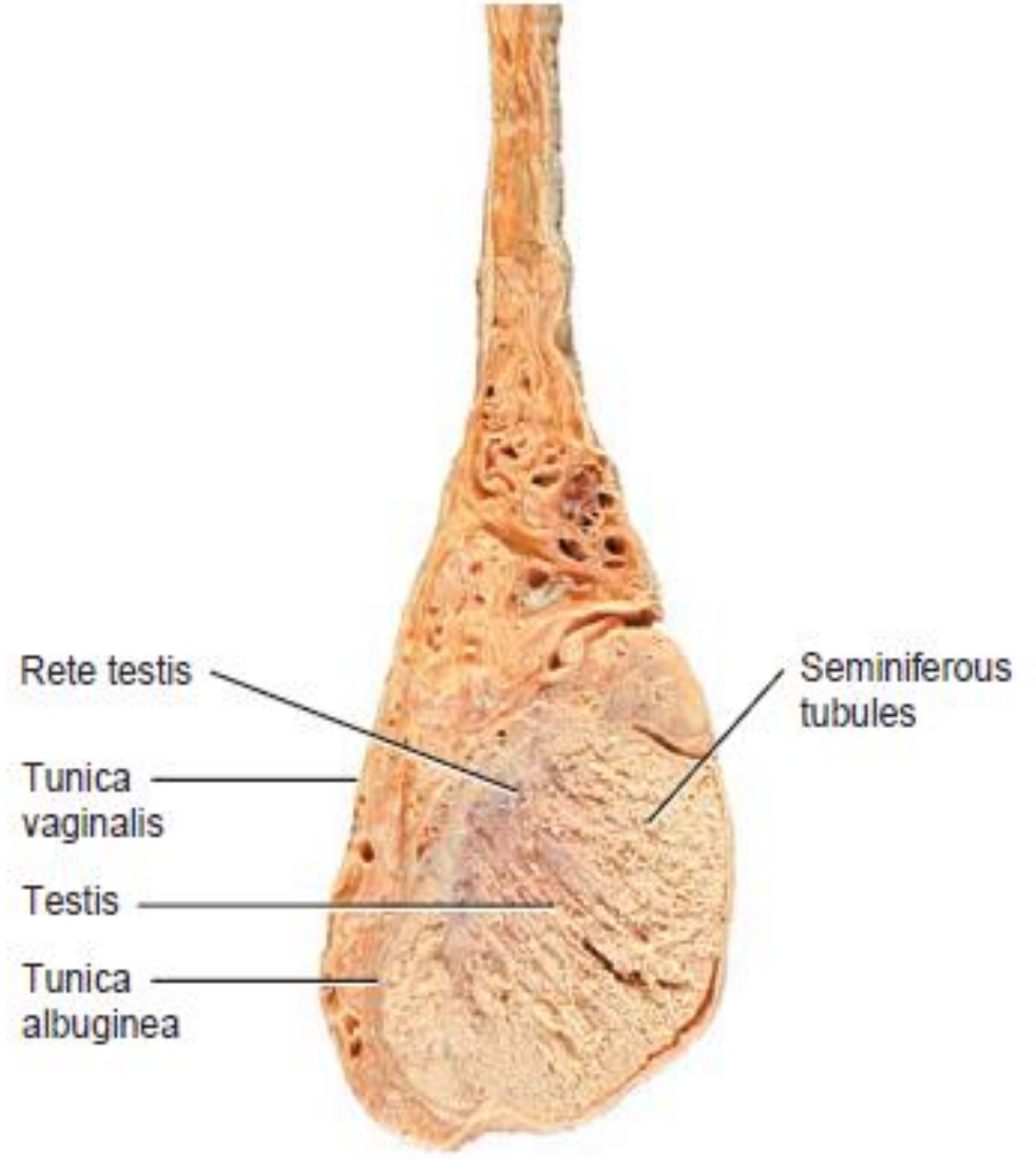
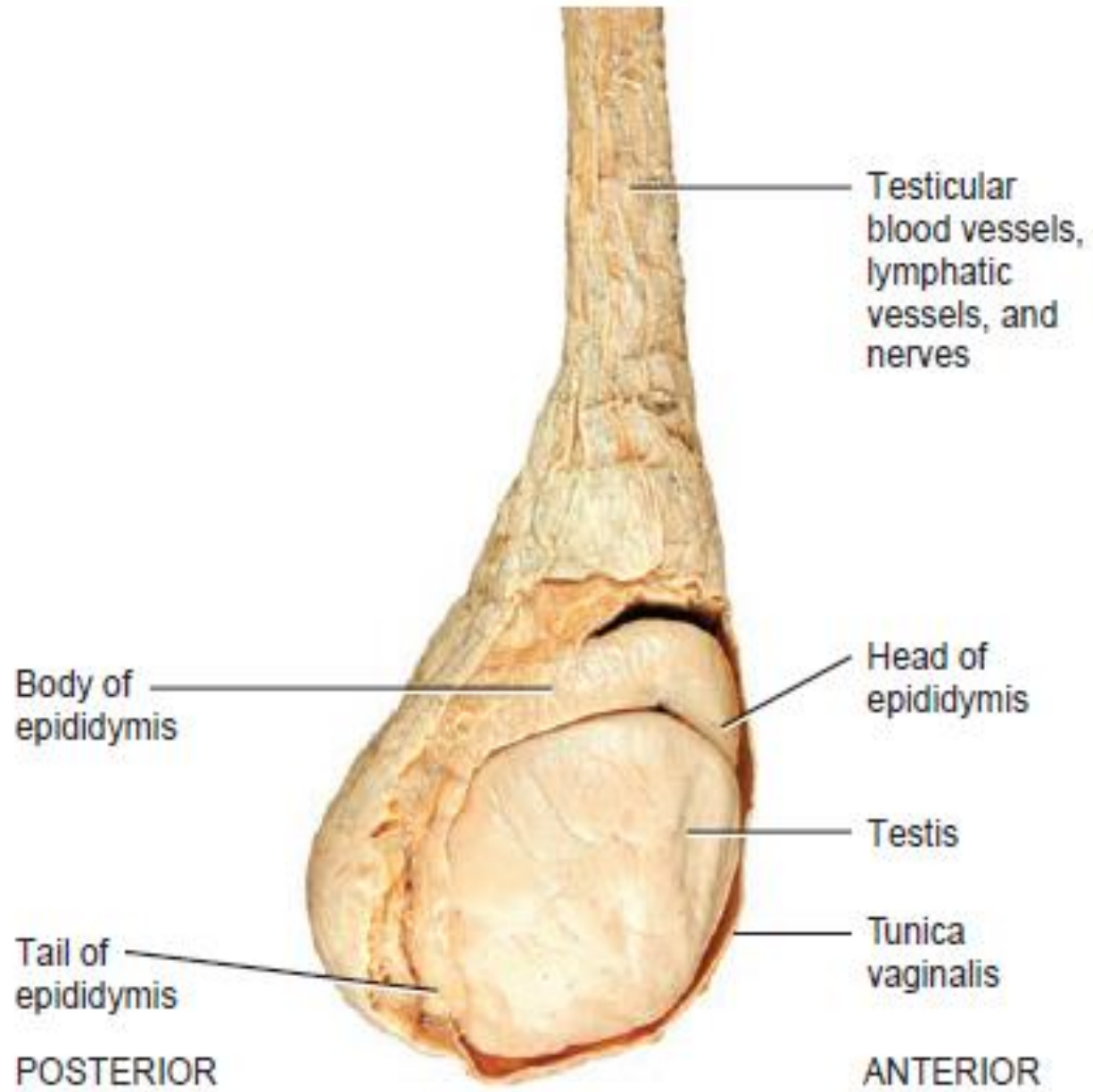
Normal Structure

- Scrotal sac include the testicle and epididymis along with lower end of the spermatic cord and the tunica vaginalis that forms the outer serous investing layer
- Structurally, the main components of the testicle are the seminiferous tubules
- Histologically, the seminiferous tubules are formed of a lamellar connective tissue membrane and contain several layers of cells





SUPERIOR



CONT'D

- In the adult, the cells lining the seminiferous tubules are of 2 types:
- 1. **Spermatogonia** or germ cells which produce spermatocytes (primary and secondary), spermatids and mature spermatozoa.
- 2. **Sertoli cells** which are larger and act as supportive cells to germ cells, also produces ABP (Androgen Binding Protein)
- The fibrovascular stroma present between the seminiferous tubules contains varying number of **interstitial cells of Leydig**
- The main source of testosterone and other androgenic hormones in males
- Main functions of the testis are to produce sperms and testosterone

CONGENITAL ANOMALIES OF THE PENIS

Hypospadias and Epispadias

- Malformation of the urethral groove (meatus) and urethral canal may create abnormal openings
 - either on the ventral surface of the penis (hypospadias) – more common
 - or on the dorsal surface (epispadias).

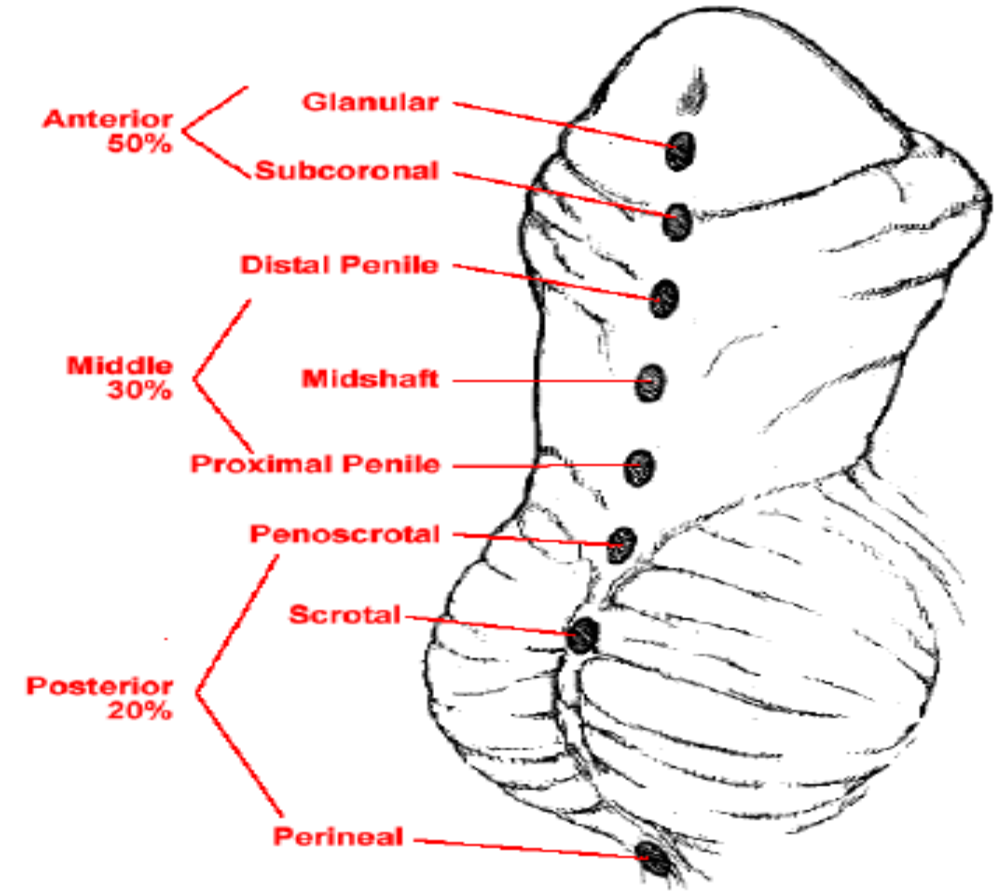
1. Hypospadias

- Urethra opening on **ventral surface** of penis
- Occurs in **1** in every **300** male children.

HYPOSPADIAS CONT'D

- There are several forms of hypospadias, classified according to location:
 - **glanular** - opening on the proximal glans penis
 - **coronal** - opening at the coronal sulcus
 - **penile shaft** – opening along the shaft
 - **penoscrotal** – opening near the beginning of the scrotum
 - **perineal** – opening at the back of the scrotum
- The constricted abnormal opening results in urinary tract obstruction and ascending urinary tract infections.

HYPOSPADIAS



CLINICAL PRESENTATIONS

- Difficulty directing the urinary stream and stream spraying.
- **Chordee** (curvature of the penis) causes ventral bending and bowing of the penile shaft. Chordee can prevent sexual intercourse; a penis that is straight when erect is necessary for satisfactory sexual function.
- Perineal or penoscrotal hypospadias necessitates urinating in the **sitting position**.
- Abnormal (hooded) appearance of the penis, caused by deficient or absent ventral foreskin.
- Concomitant anomalies
 - Any degree of hypospadias is an expression of **feminization**
 - Treatment is surgical correction

Characteristics

Characteristics

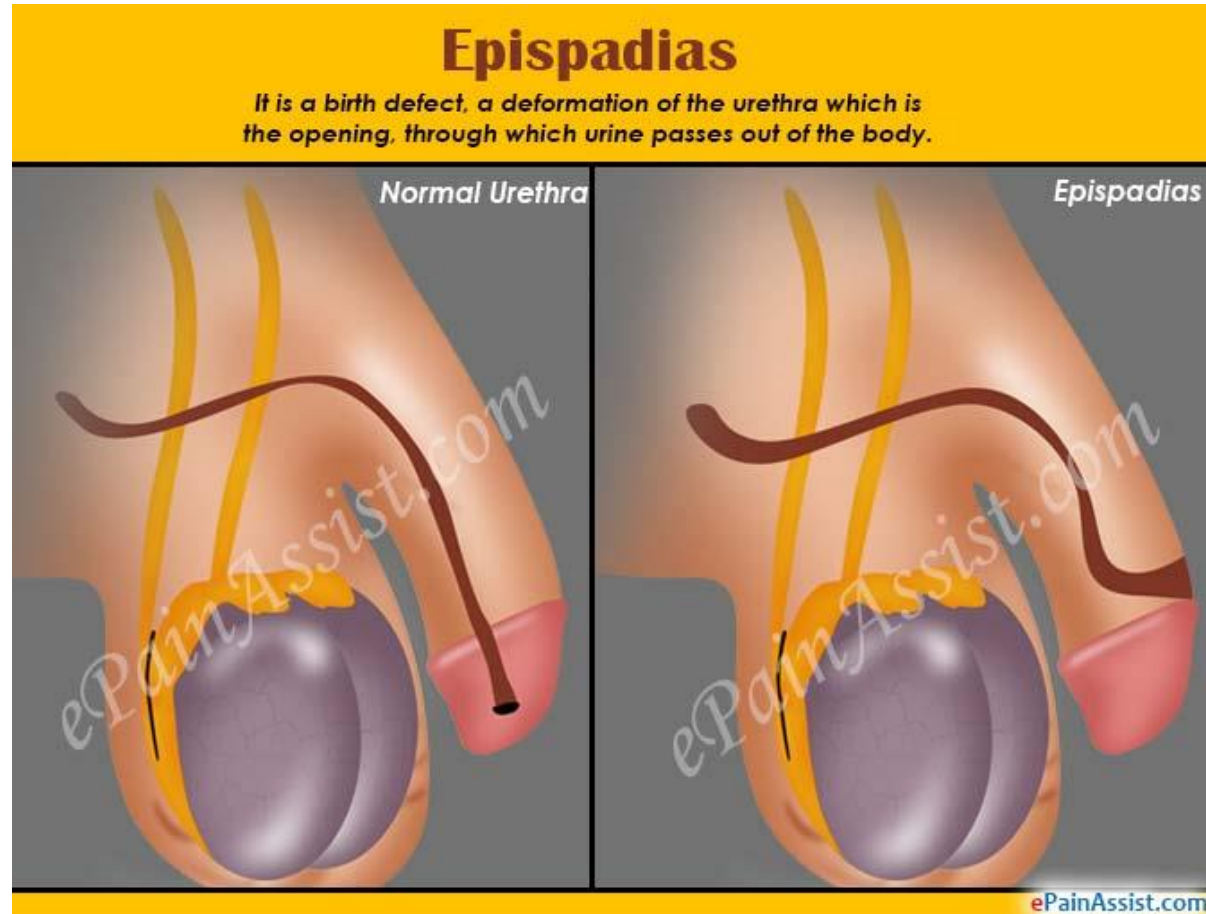
- Dorsal Hooded Prepuce
- Ventral Chordee
- Ectopic meatus



2. EPISPADIAS

- Urethra opening on **dorsal surface** of penis
- May be associated with **cryptorchidism** or malformations of urinary tract
- Abnormal opening may be constricted and **predispose to UTI**
- May cause **infertility** due to impairment of ejaculation
- Treatment is corrective **surgical correction**

EPISPADIAS



PHIMOSIS

- A condition in which the contracted foreskin cannot be **retracted (pulled)** over the glans.
- This can also occur when the **orifice** of the prepuce is too small to permit its **normal retraction**.

Causes

- May be **congenital** due to developmental anomaly or **acquired**
- Chronic infection from **poor local hygiene** → scarring of the preputial ring.
- Most cases occur in **uncircumcised males**.
- Phimosis can also occur if the foreskin is forced back before it is ready.



NORMAL



Type 1



Type 2

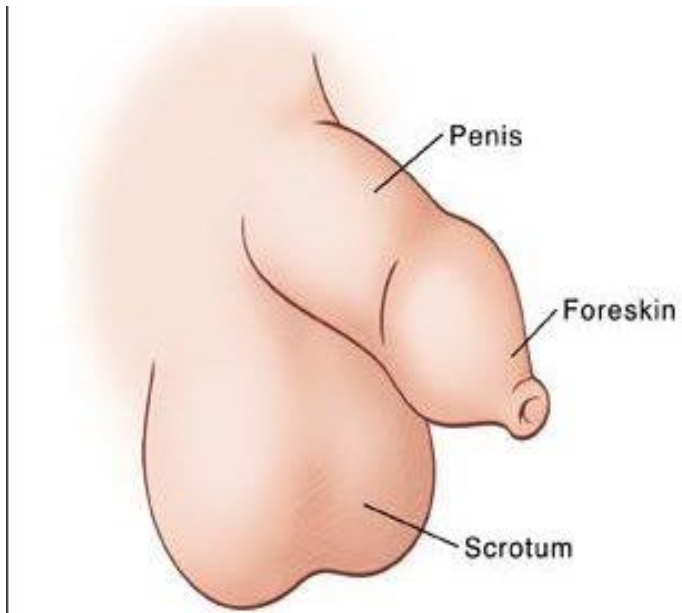


Type 3

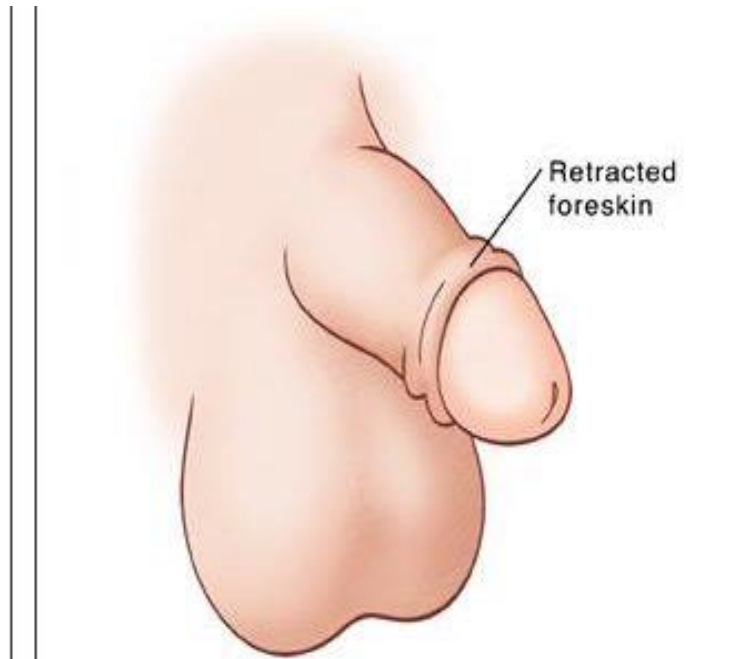


Type 4

PHIMOSIS AND PARAPHIMOSIS



Foreskin covers the uncircumcised penis.



The foreskin of an infant or young child

PHIMOSIS



PRESENTATION

- It causes **bulging** of the foreskin during urination.
- **Edema**, **erythema**, and **tenderness** of the prepuce and the presence of purulent discharge usually cause the patient to seek medical attention.
- Inability to retract the foreskin is a less common complaint

PARAPHIMOSIS

- Occurs when the foreskin is **retracted behind the corona** (or crown) of the penis and cannot be returned to the unretracted position.
- This can cause **entrapment** of the penis and **impairing** the drainage of blood.

Causes

- This is due to chronic inflammation under the redundant foreskin, which leads to contracture of the preputial opening (phimosis) and formation of a tight ring of skin when the foreskin is retracted behind the glans.
- Iatrogenic causes: during penile examination or penile cleaning.

PATHOLOGY

- The skin ring causes **venous congestion** leading to **edema** and enlargement of the glans, which make the condition worse



Phimosis



**Phimosis
with
inflammation**



**Adherent
foreskin**



Paraphimosis

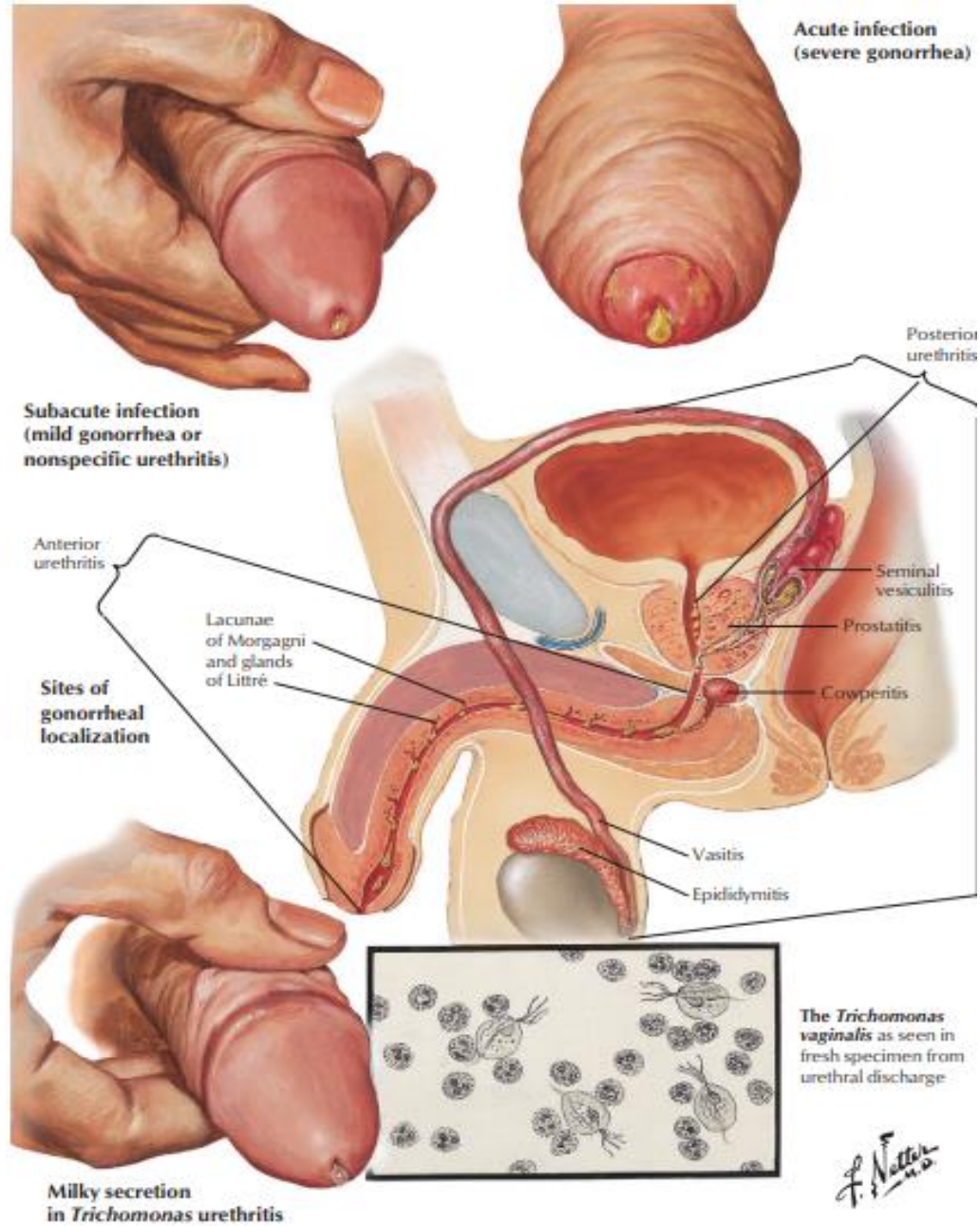


Strangulation by metal ring

W. J. Nutter

INFLAMMATORY (INFECTIOUS) CONDITIONS OF THE PENIS

- **Gonococcal urethritis** – causes a **purulent urethral discharge** → may lead to scarring & development of **urethral stricture**
- **Non gonococcal urethritis** – causes a purulent discharge → development of **Reiter's syndrome**
- **Balanoposthitis** is the term used for non-specific inflammation of the inner surface of the prepuce (**balanitis**) and adjacent surface of the glans (**posthitis**).
- Other common infections are sexually transmitted infections such as **syphilis**, **herpes genitalis**, **chancroid**, **granuloma inguinale** (*Calymmatobacterium granulomatis*), **lymphomagranuloma venereum**





Chancre of
coronal sulcus



Chancre
of glans

Multiple chancres
(shaft and meatus)



Penoscrotal
chancre



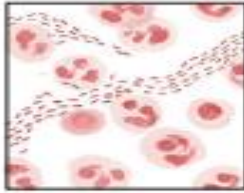
Spirochetes in dark
field examination

F. Netter M.D.

Soft chancre
of chancroid



Bacillus
ducrey



Chancroid under foreskin with marked adenitis



Lymphogranuloma
venereum



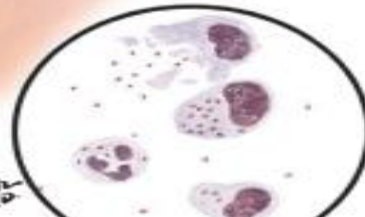
Positive
Frei test



Granuloma
inguinale



Lesions involving
groin, scrotum,
and penis



Donovan bodies in
inflammatory cells

TUMOURS OF THE PENIS

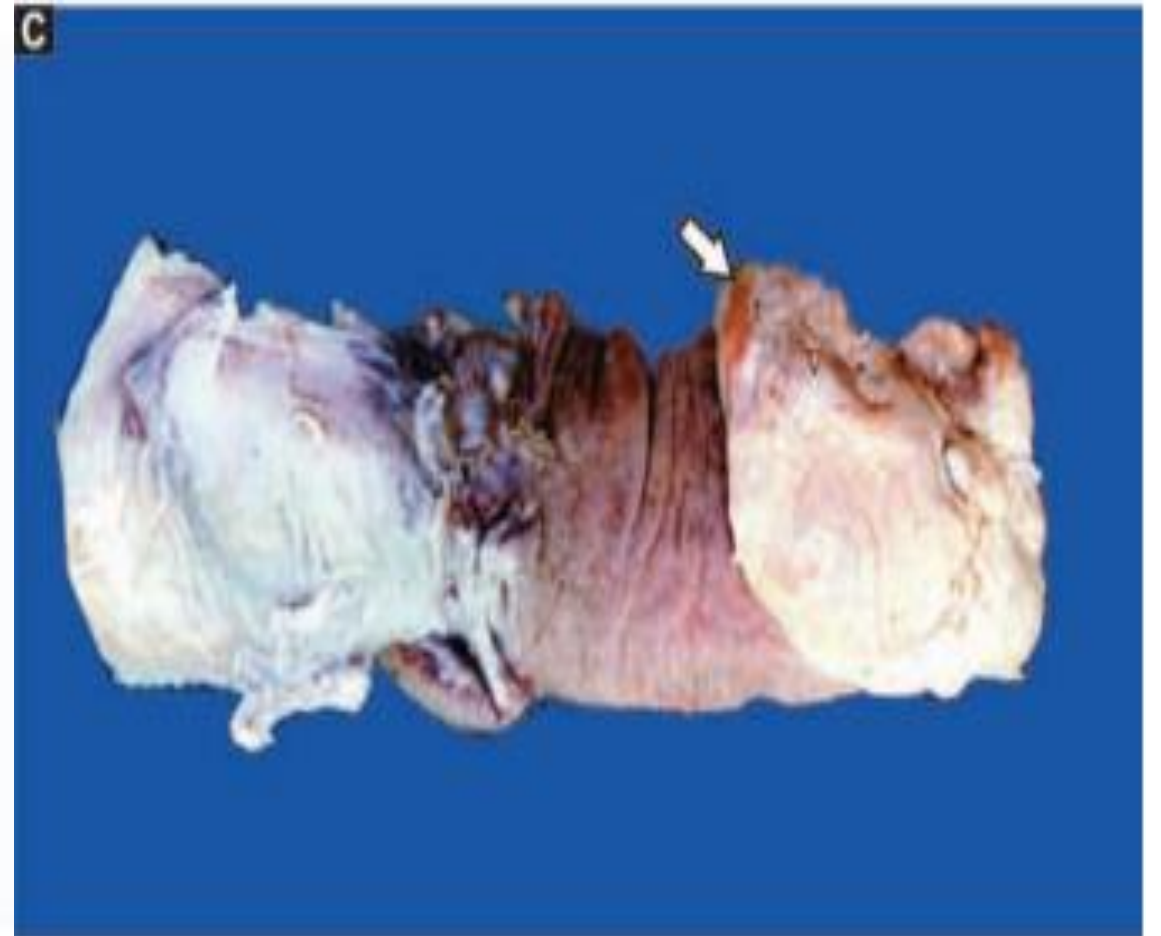
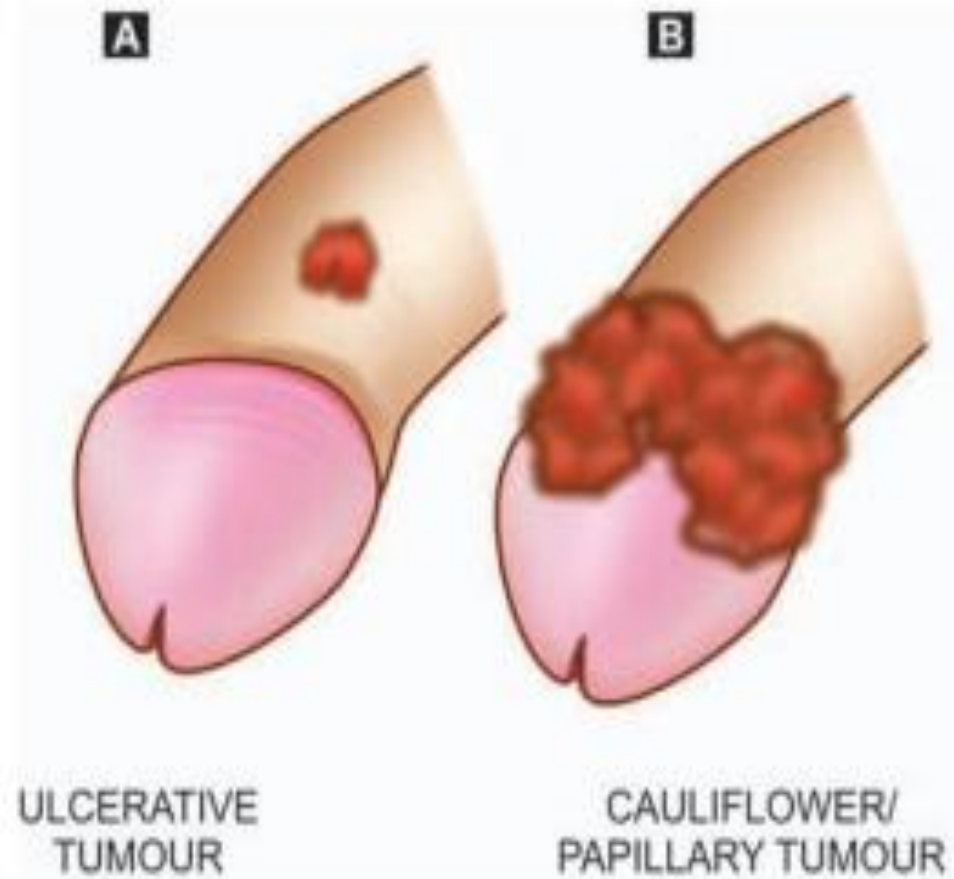
Benign tumour (Condyloma Acuminatum)

- Sexually transmitted tumour caused by HPV **type 6** less common **type 11**
- Related to the **common wart** & occurs on moist mucocutaneous surface – coronal sulcus & perianal area

Morphology

- Consist of single or multiple sessile/pedunculated, red papillary excrescence, exophytic growth pattern
- Histologically as **branching villous connective** tissue stroma → superficial hyperkeratosis and acanthosis

Condyloma Acuminatum



TUMOURS OF THE PENIS – CONT'D

Pre-Malignant tumours

Carcinoma in Situ (CIS)

- **Bowen disease** – appears as solitary, thickened, graywhite, opaque plaque.
- Manifests on the glans or prepuces as single or multiple shiny red/velvety plaques
- Histologically the cells show marked dysplasia with large **hyperchromatic nuclei** & lack **orderly** maturation
- In about 10% of patients it transforms in **squamous cell carcinoma**

Bowen disease



TUMOURS OF THE PENIS – CONT'D

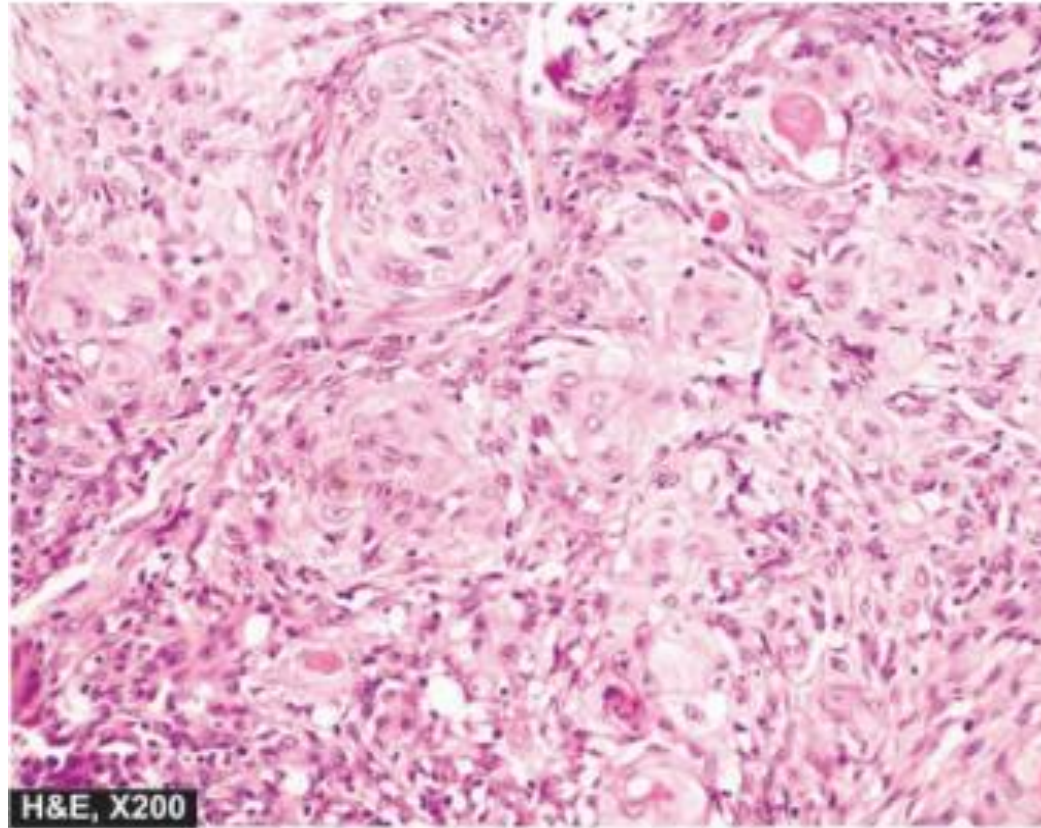
- *Bowenoid papulosis* – appears as multiple reddish brown popular lesions
- Histologically not different from bowen disease but it never develops in to invasive carcinomas
- Histologically there is orderly maturation of epithelial cells in hyperplastic epidermis with scattered hyperchromatic nuclei and dysplastic cells
- Linked to the presence of HPV type 16

TUMOURS OF THE PENIS – CONT'D

Invasive carcinoma

- **Squamous cell carcinoma** is extremely rare and is common in uncircumcised population
- Is linked with **HPV 16 & 18** and affects ages between **40 – 70 yrs**
- Morphologically can be **papillary** or **flat** & commonly starts from the glans or the sulcus & with progression can **become ulcerated**
- Histologically appears as squamous cells carcinomas with fungating and ulcerating varying **degree of differentiation**
- Rarely metastasize

Histology of the carcinoma of the penis

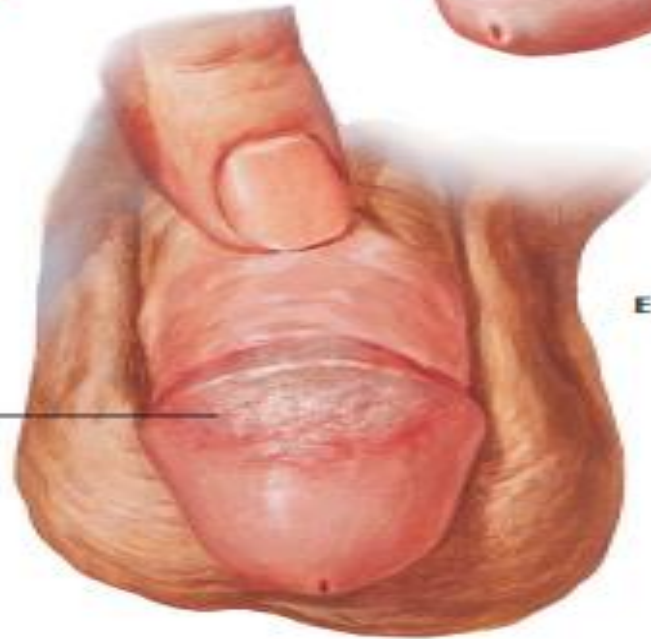




Venereal warts

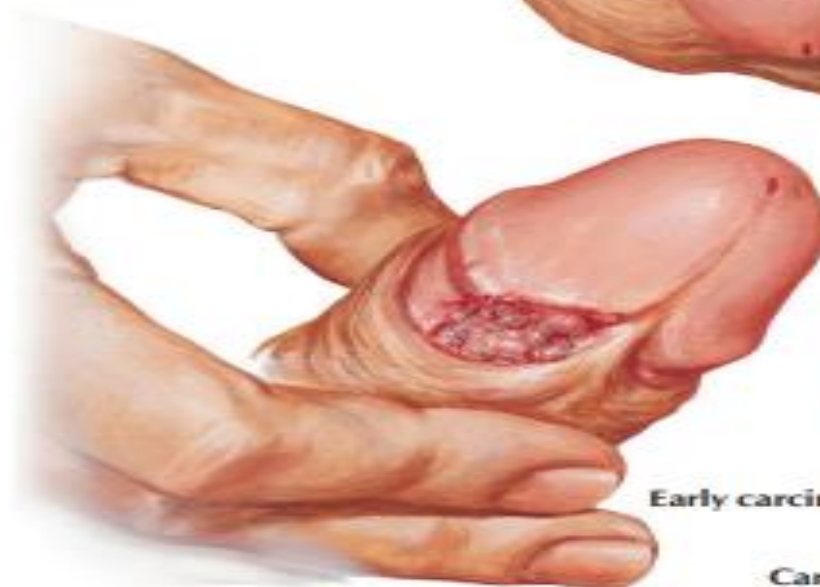


Leukoplakia

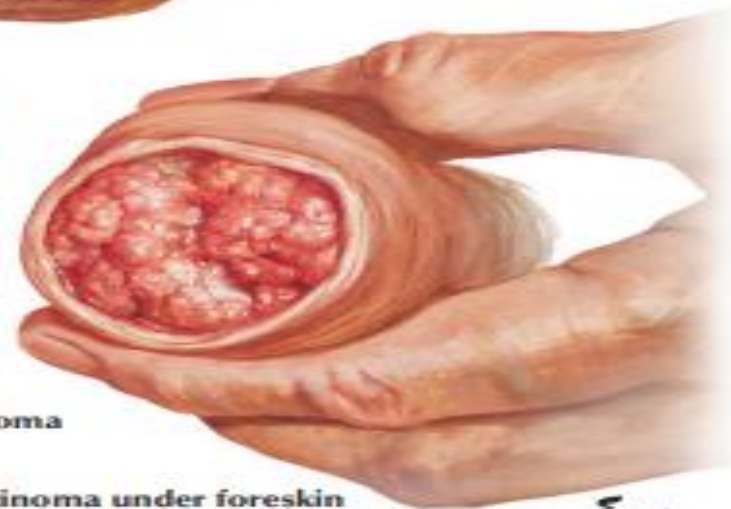


Erythroplasia of Queyrat

Lesion



Early carcinoma



Carcinoma under foreskin

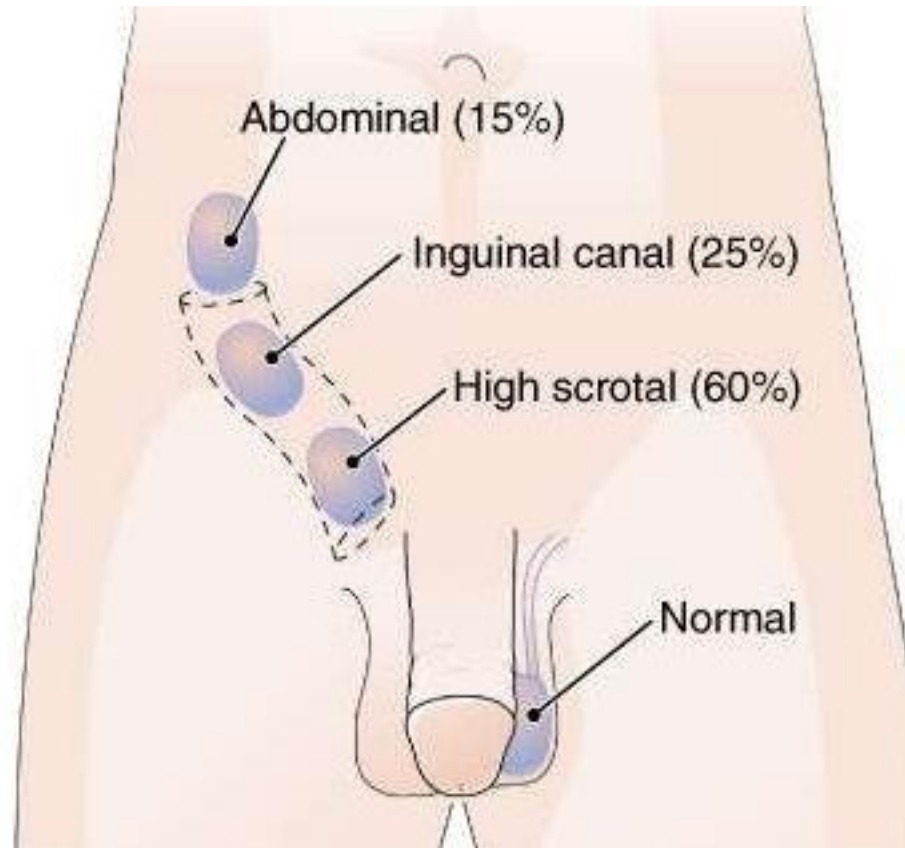
W. H. W. H. W.

TESTIS, EPIDIDYMISS AND PARATESTICULAR TISSUES

CRYPTORCHIDISM

- Permanent **retention of testes** outside scrotum
- Occurs in about 1% boys
- Bilateral in 25% cases
- Most retained testes descend in scrotum within first year
- Undescended testes are retained in **abdominal cavity (25%)** or **inguinal area (70%)**, others (5%)
- If not descended by age 3, consider surgery

CRYPTORCHIDISM



**UNDESCENDED
TESTICLE**

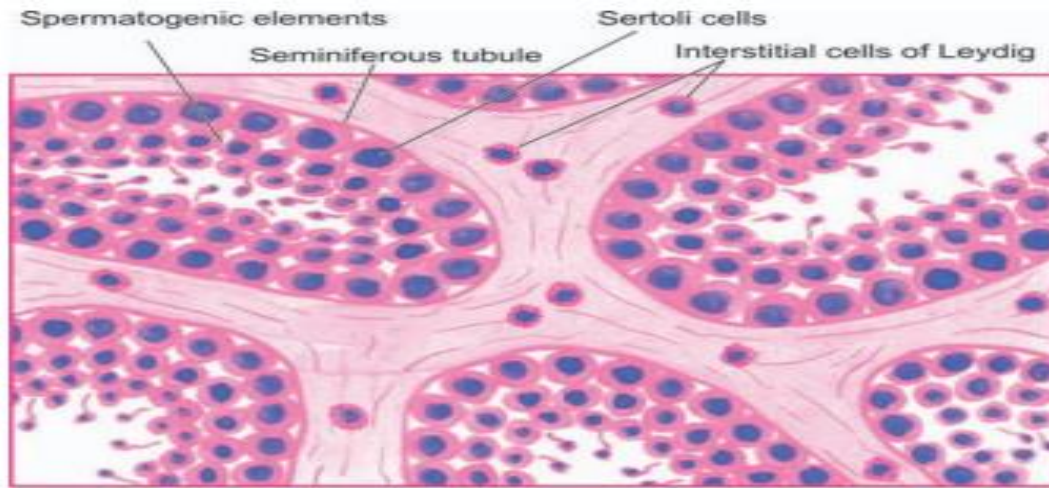
ETIOLOGY

- **Not known** in majority of cases. However, a few apparent causes associated with cryptorchidism are as under:
- 1. **Mechanical factors** e.g. short spermatic cord, narrow inguinal canal, adhesions to the peritoneum.
- 2. **Genetic factors** e.g. trisomy 13, mal-development of the scrotum or cremaster muscles.
- 3. **Hormonal factors** e.g. deficient androgenic secretions.

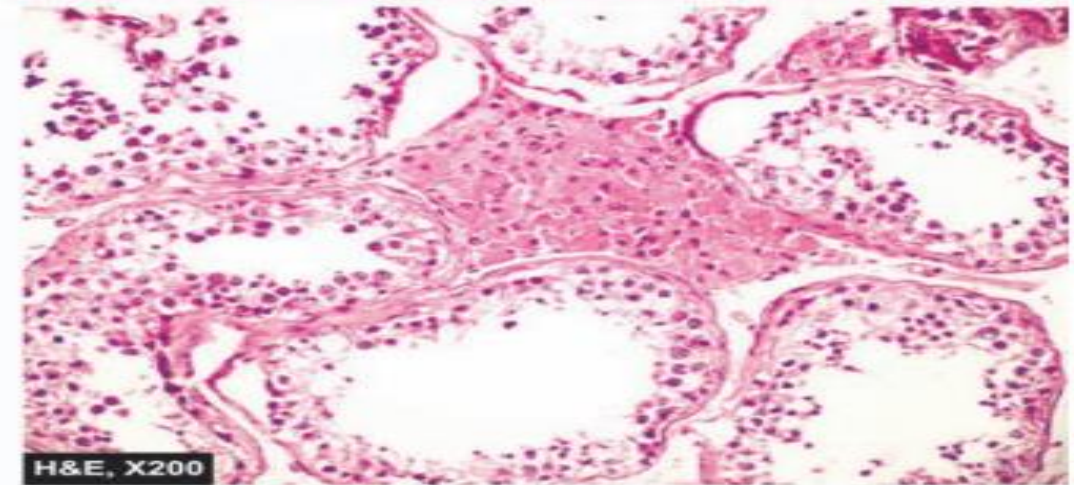
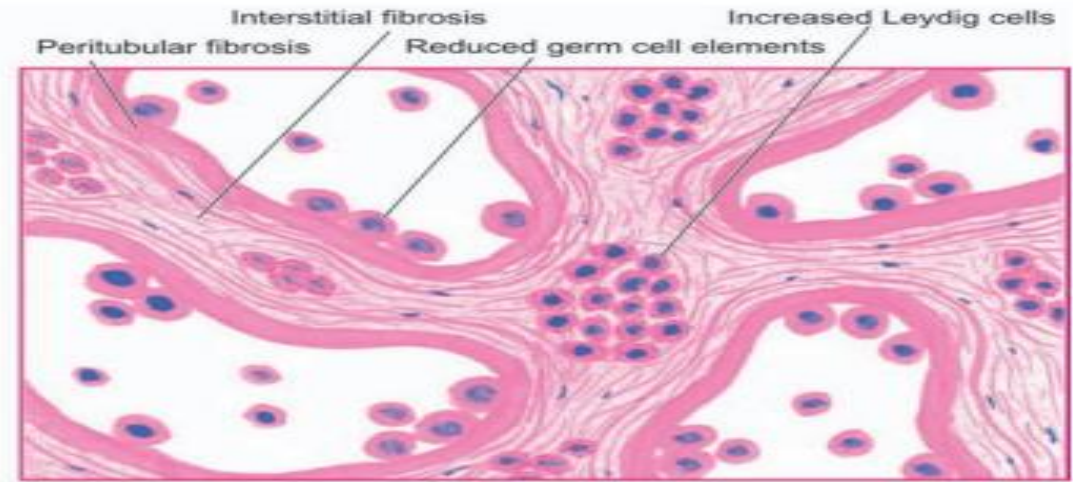
MORPHOLOGY

- Cryptorchid testes in adults are **small** and **brown**
- Tubules are atrophic, basement membranes thickened
- Arrest in development of **germ cells** → Hyalinization
- ↑interstitial fibrovascular stroma & conspicuous presence
Leydig cells

HISTOLOGY OF THE TUBULES



A, NORMAL TESTIS



B, CRYPTORCHID TESTIS

CLINICAL FEATURES & COMPLICATION

- Cryptorchidism is completely asymptomatic and is discovered only on physical examination
- **Sterility-infertility**. Bilateral cryptorchidism is associated with sterility while unilateral disease may result in infertility.
- Infertility in most cases due to ↑ intra-scrotal temperature
- **Inguinal hernia**. A concomitant inguinal hernia is frequently present along with cryptorchidism
- **Malignancy**, 40x risk than in normal testis – commonly a seminoma & embryonal carcinoma

CAUSES OF MALE INFERTILITY (SUMMARY)

PRE-TESTICULAR CAUSES:

- 1. **Hypopituitarism**. Pre-pubertal or post-pubertal hypopituitarism such as from tumour, trauma, infarction, cyst and genetic deficiency of FSH and LH secretion.
- 2. **Oestrogen excess**. Endogenous excess such as from hepatic cirrhosis, adrenal tumour, Sertoli and Leydig cell tumour; or exogenous excess such as in the treatment of carcinoma of the prostate.
- 3. **Glucocorticoid excess**. Endogenous excess may occur in Cushing's syndrome while exogenous excess may occur in the treatment of ulcerative colitis, bronchial asthma, rheumatoid arthritis etc.
- 4. **Other endocrine disorders**. Hypothyroidism and diabetes mellitus are associated with hypospermatogenesis.

CAUSES OF MALE INFERTILITY (SUMMARY) – CONT'D

TESTICULAR CAUSES:

- 1. **Agonadism** i.e. total absence of the testes.
- 2. **Cryptorchidism** or undescended testis described above.
- 3. **Maturation arrest** i.e. failure of spermatogenesis beyond one of the immature stages.
- 4. **Hypospermatogenesis** i.e. presence of all the maturation stages of spermatogenesis but in decreased number.
- 5. **Sertoli cell-only syndrome**. Congenital or acquired absence of all germ cells so that the seminiferous tubules are lined by Sertoli cells only.
- 6. **Klinefelter's syndrome**. An XXY intersexuality characterised by primary hypogonadism, azoospermia, gynaecomastia, eunuchoid built and subnormal intelligence.
- 7. **Mumps orchitis** occurring as a complication of parotitis .
- 8. **Irradiation damage** resulting in permanent germ cell destruction.

CAUSES OF MALE INFERTILITY (SUMMARY) – CONT'D

POST-TESTICULAR CAUSES:

- 1. **Congenital block** e.g. absence or atresia of vas deferens.
- 2. **Acquired block** e.g. due to gonorrhoea and surgical intervention.
- 3. **Impaired sperm motility** in the presence of normal sperm counts e.g. immotile cilia syndrome

ATROPHY OF TESTIS

Is a cause of infertility

- Cryptorchidism
- Mumps – when infection occurs at or after puberty
- Liver cirrhosis
- Oestrogens or gonadotropin-releasing hormone analogues used in treatment of prostate cancer
- Radiation exposure
- Chemotherapy, environmental toxins
- Testicular regression syndrome

INFLAMMATORY CONDITIONS

- Inflammation of the testis is termed as **orchitis** and of epididymis is called as **epididymitis**
- Common routes of spread of infection are via the vas deferens, or via lymphatic and haematogenous routes
- Infection is caused by **urethritis**, **cystitis**, **prostatitis** and **seminal vesiculitis**
- Other causes are **mumps**, **smallpox**, **dengue fever**, **influenza**, **pneumonia** and **filariasis**
- STIs (*Neisseria gonorrhoeae* and *Chlamydia trachomatis*)
- UTIs (*Escherichia coli* and *Pseudomonas*)

CONT'D

Morphology

- Grossly, in acute stage the testicle is firm, tense, swollen and congested
- In chronic cases, there is usually variable degree of atrophy and fibrosis
- Histologically, acute orchitis and epididymitis are characterised by **congestion**, **oedema** and diffuse infiltration by **neutrophils**, **lymphocytes**, **plasma cells** and **macrophages** or formation of neutrophilic abscesses
- In chronic epididymo-orchitis, there is focal or diffuse chronic inflammation, disappearance of **seminiferous tubules**, **fibrous scarring** and destruction of **interstitial Leydig cells**

INFLAMMATORY CONDITIONS

Granulomatous (autoimmune) orchitis

- Solid nodular enlargement of testis, may mimic a tumour
- **Non-caseating granulomas** present within tubules and connective tissue
- Granulomatous response partly due to products of disintegrated sperm

Tuberculous Epididymo-orchitis

- Usually begins in **epididymis** and spreads to **testis**
- **Classic granulomas** seen

INFLAMMATORY CONDITIONS – CONT'D

Viral Orchitis

- Caused by **mumps virus** infection after puberty
- Is unilateral & associated with tender enlargement of the testis
- Inflammatory infiltrate is composed of mainly lymphocytes & plasma cells
- It can lead to **sterility**.

VASCULAR DISORDERS

Torsion

- When the testis rotates on its vascular pedicle interrupting blood sup.
- A **urological emergency** common in boys and young men
- Sudden muscular effort or physical trauma or bell-Clapper derformity
- Characterized by vascular engorgement & **ischaemia develops** → scrotal swelling & pain
- Congestion to wide spread haemorrhage → haemorrhagic testicular infarction – coagulative necrosis
- Testis markedly enlarged → sac of soft, necrotic haemorrhage tissue

VASCULAR DISORDERS – CONT'D

Hydrocele

- Pathological accumulation of Serous **fluid** in the scrotum
- Result of fluid accumulation in the **cavity** bounded by tunica vaginalis
- Causes are **trauma**, **systemic oedema**, such as in **cardiac failure** and **renal disease**, and as a complication of **gonorrhoea**, **syphilis** and **tuberculosis**.

Haematocele

- Haematocele is **haemorrhage** into the sac of the tunica vaginalis. It may result from **direct trauma**, from injury to a **vein** by the needle, or from **haemorrhagic** diseases.

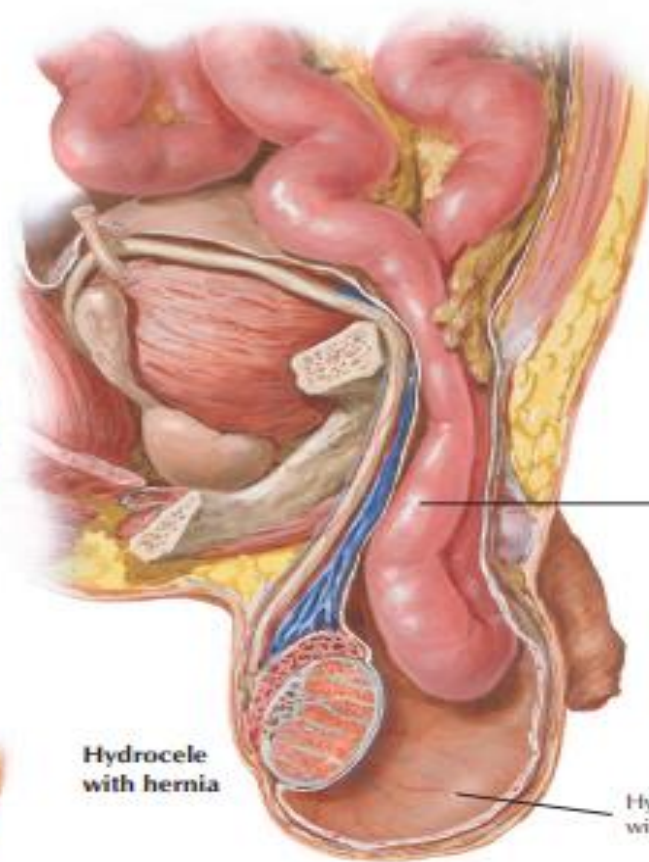
Hydrocele



Bladder

Testis

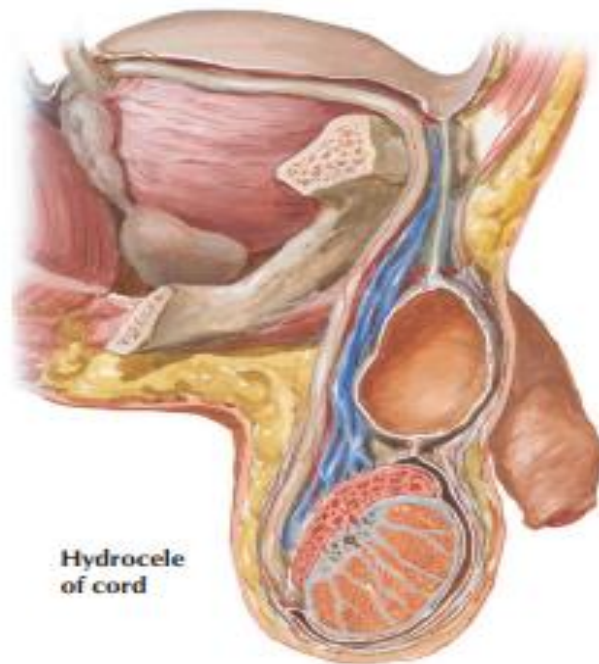
Simple hydrocele



Herniated intestine

Hydrocele with hernia

Hydrocele within scrotum

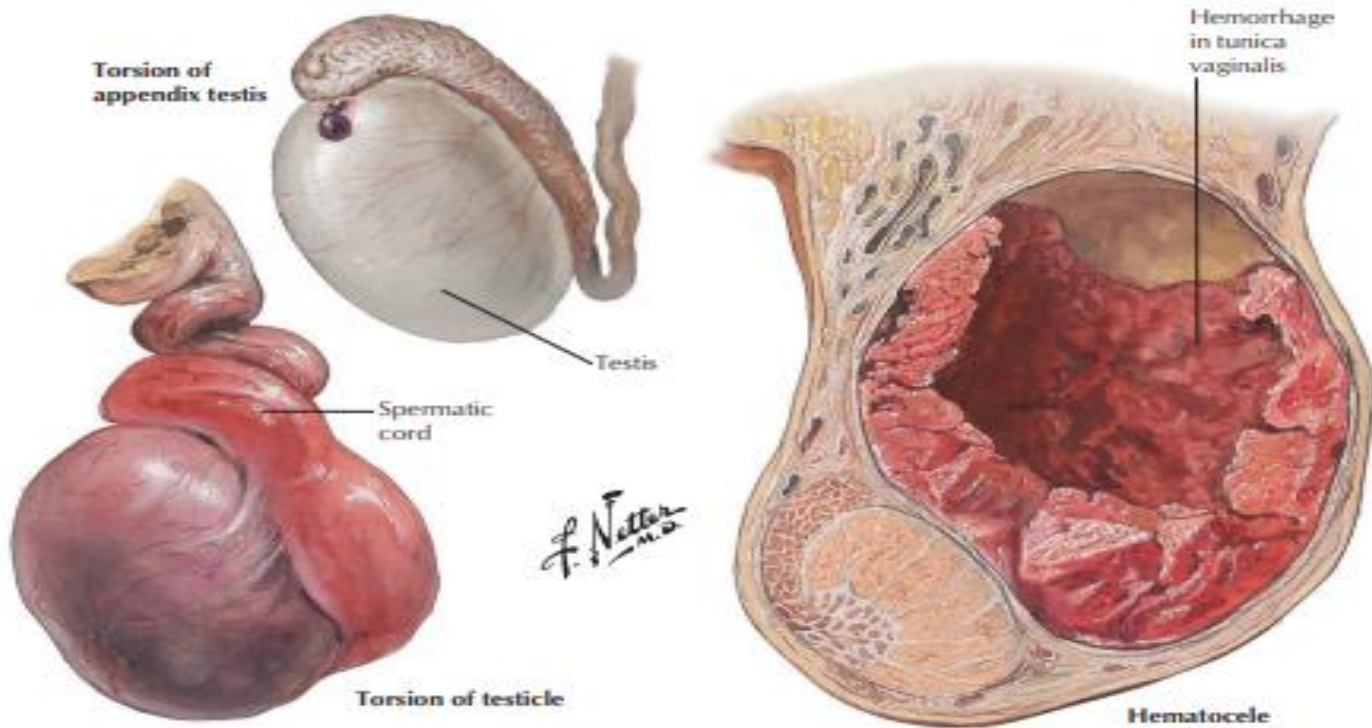
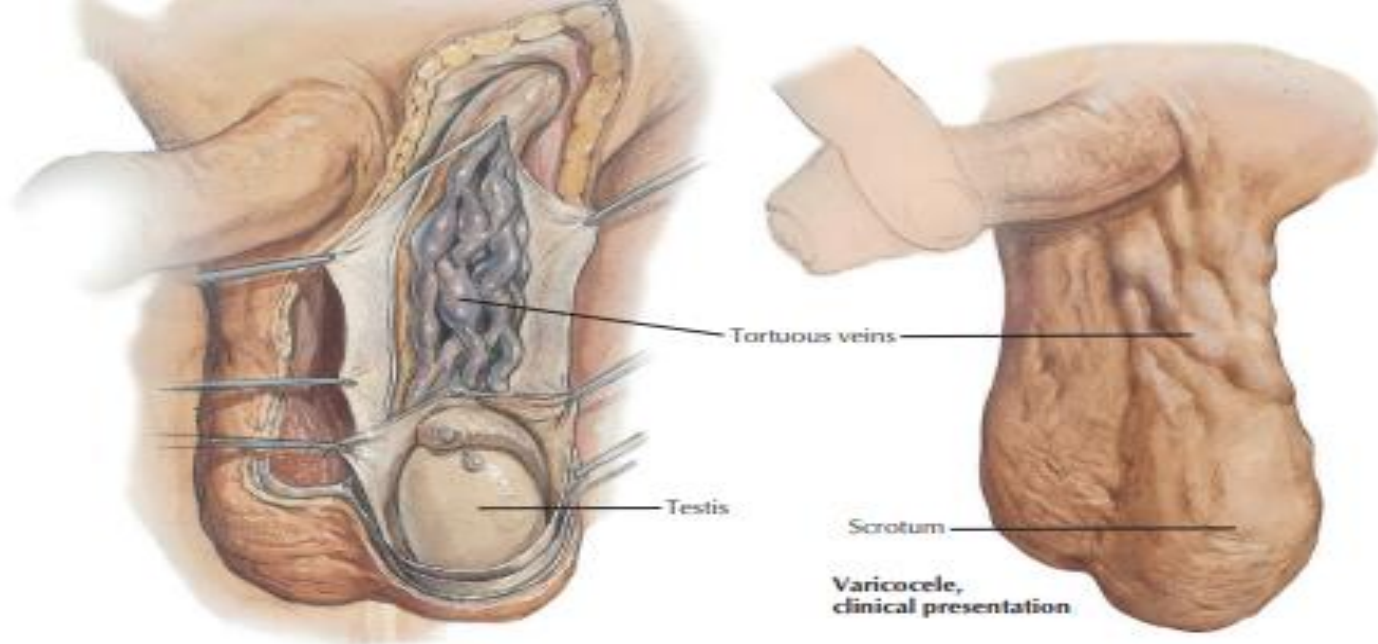


Hydrocele of cord



Spermatocele

F. Netter M.D.



VASCULAR DISORDERS – CONT'D

Varicocele

- is the **dilatation**, **elongation** and **tortuosity** of the veins of the pampiniform plexus in the spermatic cord
- Swelling of the scrotum due to the accumulation of **blood** in the *pamphiform plexus* in the spermatic cord
- Due to increase in the back flow of pressure in the veins
- **Primary or idiopathic** form is more frequent and is common in young unmarried men
- **Secondary form** occurs due to pressure on the spermatic vein by enlarged liver, spleen or kidney. It is commoner in middle-aged people

TESTICULAR TUMOURS

GERM CELL TUMOURS

- Seminoma, Spermatocytic seminoma, embryonal carcinoma, yolk sac tumour, polyembryoma, choriocarcinoma, teratoma

SEX CORD-STROMAL TUMOURS

- Leydig cell tumour, Sertoli cell tumour, granulosa cell tumour
- Mixed forms
- Incompletely differentiated forms

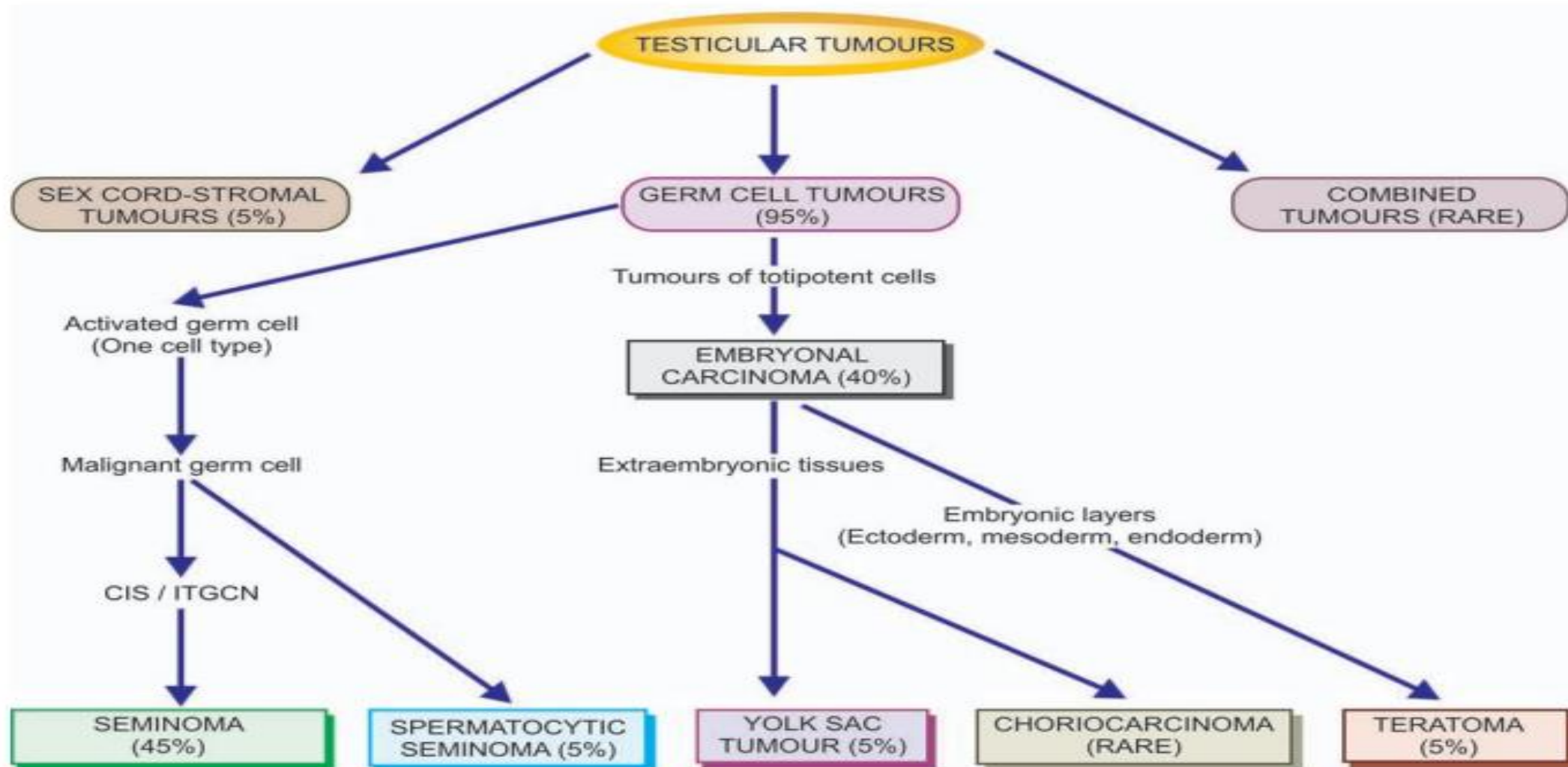
COMBINED GERM CELL-SEX CORD-STROMAL TUMOURS

- Gonadoblastoma

OTHER TUMOURS

- Malignant lymphoma (5%)
- Rare tumours

Schematic diagram showing Histogenesis of testicular tumours



GERM CELL TUMOURS

- Two major clinical categories of germ cell tumours
 - Seminoma
 - Non-seminomatous germ cell tumours (NSGCT)
- Tumours arise from germinal (seminiferous) epithelium of mature or maturing testis
- NSGCT recapitulate embryogenesis, their diff is toward one or more component of embryo
- Seminomas are regarded as end point neoplasms, incapable of diff in any direction

SEMINOMA

- Most common type of germ cell tumors making up 50%
- In the 15- to 34-year age group, they constitute the most common tumor of men and cause approximately 10% of all cancer deaths.
- Produce bulky masses, sometimes ten times the size of the normal testis.
- The typical seminoma has a homogeneous, gray-white, lobulated cut surface, usually devoid of hemorrhage or necrosis.
- Cryptorchidism is the most important risk factor.
- Seminoma can undergo metastasis

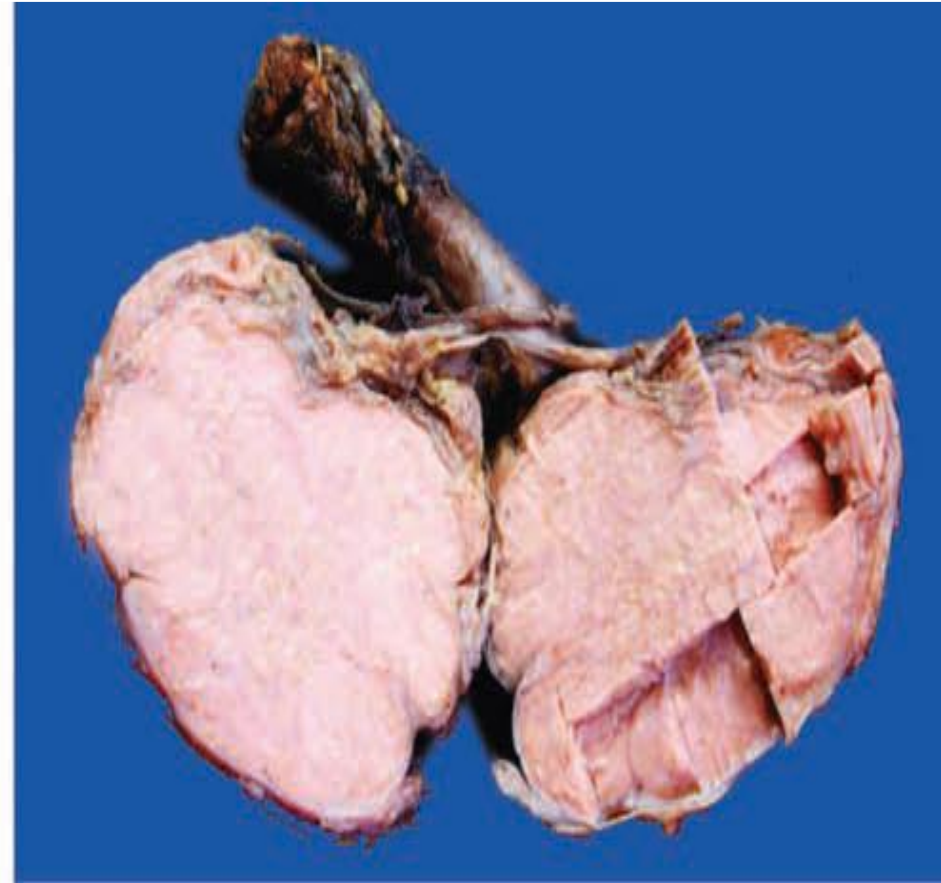
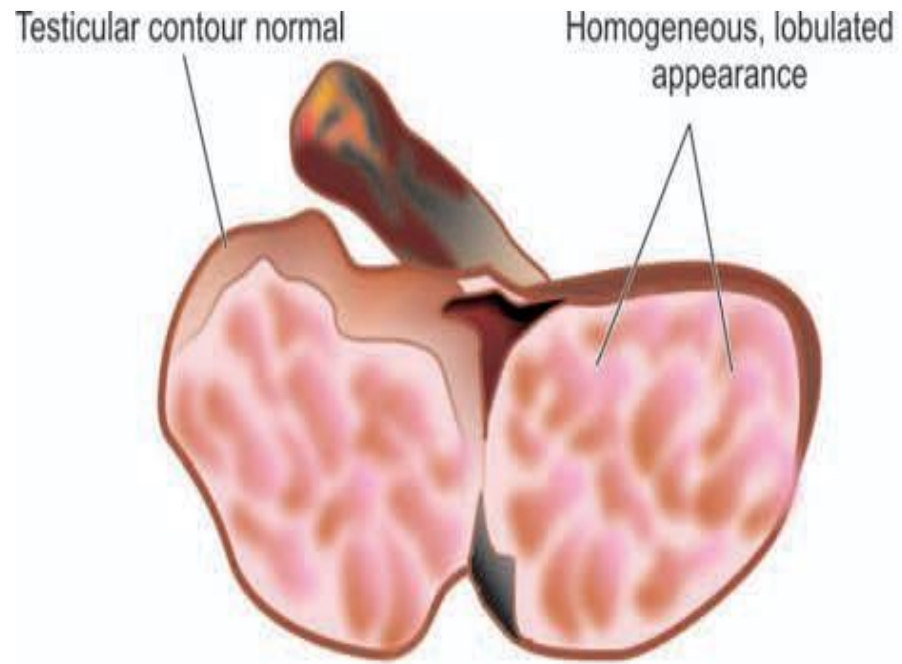
SEMINOMA

- Make up 40% of testicular tumours
- Divided into classic and spermatocytic seminoma

Classic seminoma

- Comprises 90% cases of seminomas
- Peak 40 years for classic and 65 years for spermatocytic seminoma
- Moderate size, solid, yellow
- Uniform cells, abundant clear cytoplasm arranged in nests separated by fibrous bands containing lymphocytes
- Variable amounts of glycogen within cytoplasm
- Contain PLAP

Classic seminoma



SEMINOMA – Cont'd

Spermatocytic seminoma

- Occurs in an older age group
- Soft gelatinous appearance
- Cells are pleomorphic, bizarre giant cells common, mitoses may be numerous
- No glycogen or lymphocytic infiltrate
- Reactivity for PLAP is absent
- Occurs only in testis and never seen in combination with NSGCTs
- Commonly bilateral, prognosis excellent
- Metastases practically none existent
- Simple orchiectomy is sufficient

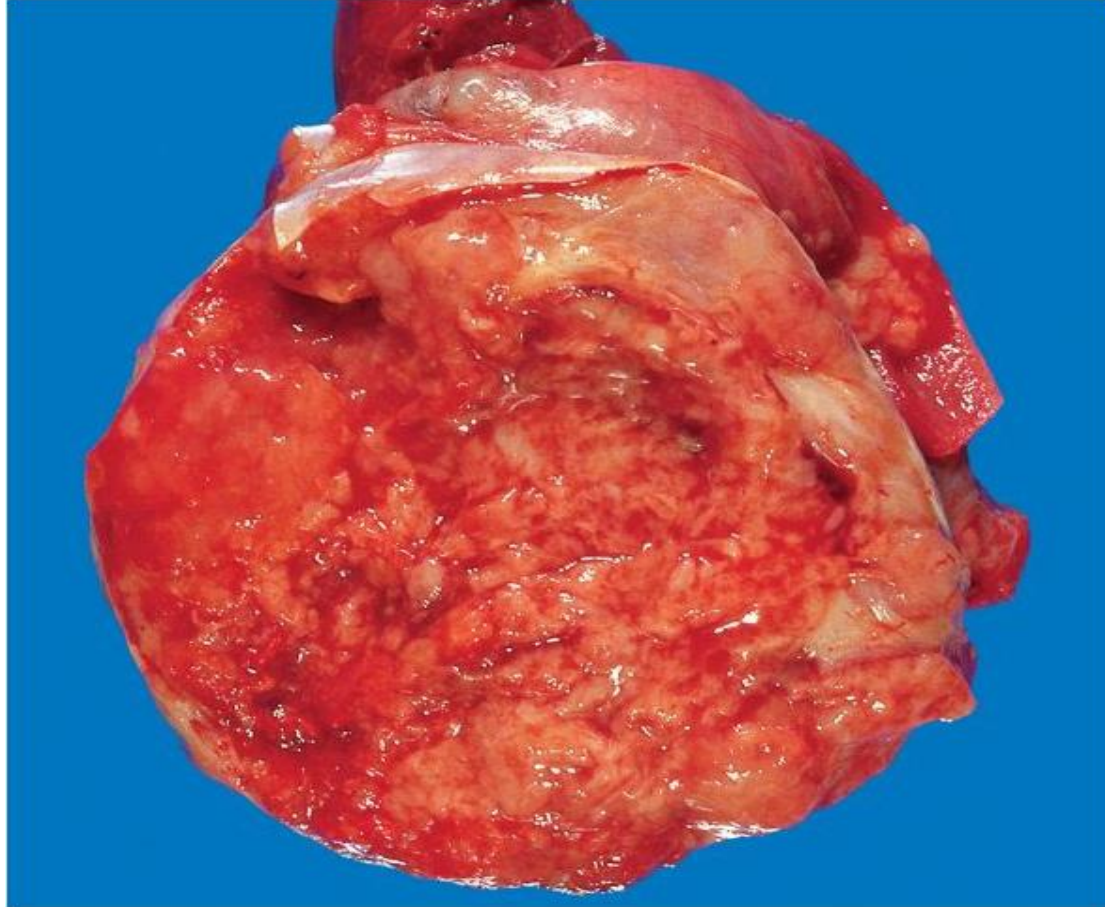
Distinguishing features btwn SGCT & NSGCT

Feature	SGCT	NSGCT
1. <i>Primary tumour</i>	Larger, confined to testis for sufficient time; testicular contour retained	Smaller, at times indistinct; testicular contour may be distorted
2. <i>Metastasis</i>	Generally to regional lymph nodes	Haematogenous spread early
3. <i>Response to radiation</i>	Radiosensitive	Radioresistant
4. <i>Serum markers</i>	hCG; generally low levels	hCG, AFP, or both; high levels
5. <i>Prognosis</i>	Better	Poor

NON-SEMINOMATOUS GERM CELL TUMOURS

- **Embryonal carcinoma** – primitive carcinoma-like cells with minimal or no signs of differentiation
- Cut surface shows variegated poorly demarcated margins punctuated by foci of haemorrhage/necrosis
- Histologically the cells grow in alveolar or tubular patterns sometimes with papillary convolutions
- 90% cases are associated with elevation of AFP or hCG or both
- is more aggressive and less radiosensitive than seminoma
- Chemotherapy is more effective in treating this tumour.

EMBRYONAL CARCINOMA



NON-SEMINOMATOUS GERM CELL TUMOURS – CONT'D

- **Yolk sac tumour** – formation of extraembryonic endoderm and mesoderm
- Is the most common testicular tissue in infants $\approx$ 3yrs
- The tumour is none capsulated, homogenous, yellow-white, mucinous appearance,
- α -fetoprotein & α -antitrypsin are elevated & can be demonstrated by immunohistochemical staining
- AFP levels are elevated in 100% cases of yolk sac tumours

NON-SEMINOMATOUS GERM CELL TUMOURS – CONT'D

- **Choriocarcinoma** – formation of well developed trophoblastic elements - ↑hCG
- Highly malignant & rare in pure form → 1%
- Morphologically do not cause testicular enlargement & are detected as small palpable nodules - <5cm
- Histologically the tumour has two cell types:
 - Syncytiotrophoblastic cell → irregular/lobular hyperchromatic nuclei
 - Cytotrophoblastic cell → regular & polygonal with distinct borders & clear cytoplasm

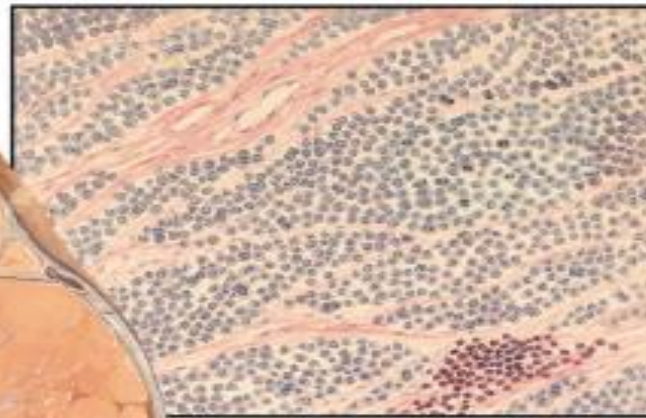
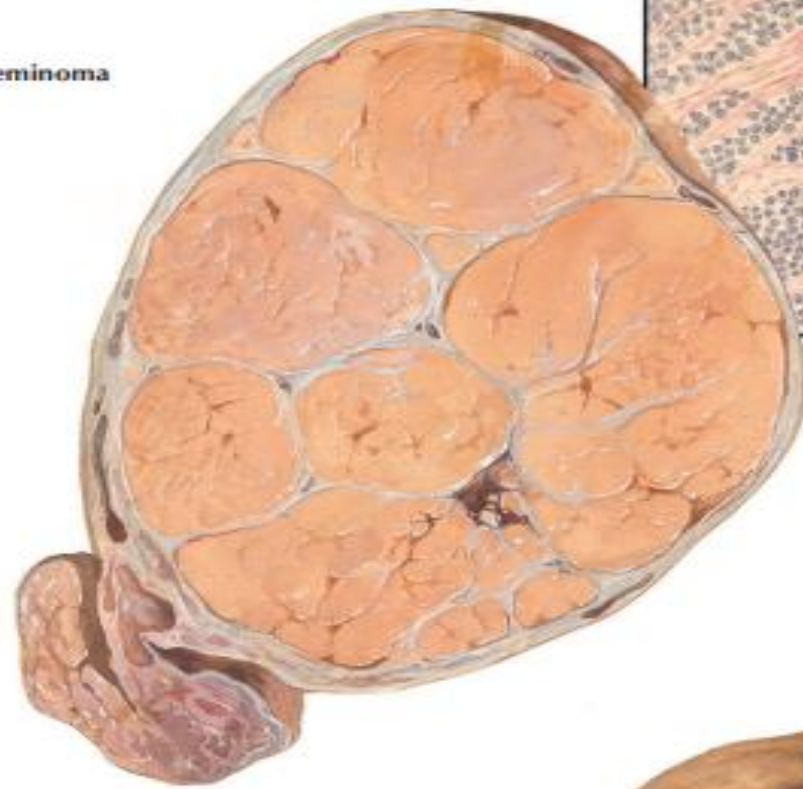
NON-SEMINOMATOUS GERM CELL TUMOURS – CONT'D

- **Teratoma** – differentiation is toward structures of embryo proper – **endodermic**, **ectodermic** and **mesodermic** tissues
- Morphologically they as a heterogeneous solid with sometimes cartilaginous & cystic areas
- Maybe;
 - 1. Mature (differentiated) teratoma
 - 2. Immature teratoma
 - 3. Teratoma with malignant transformation.

TERATOMA

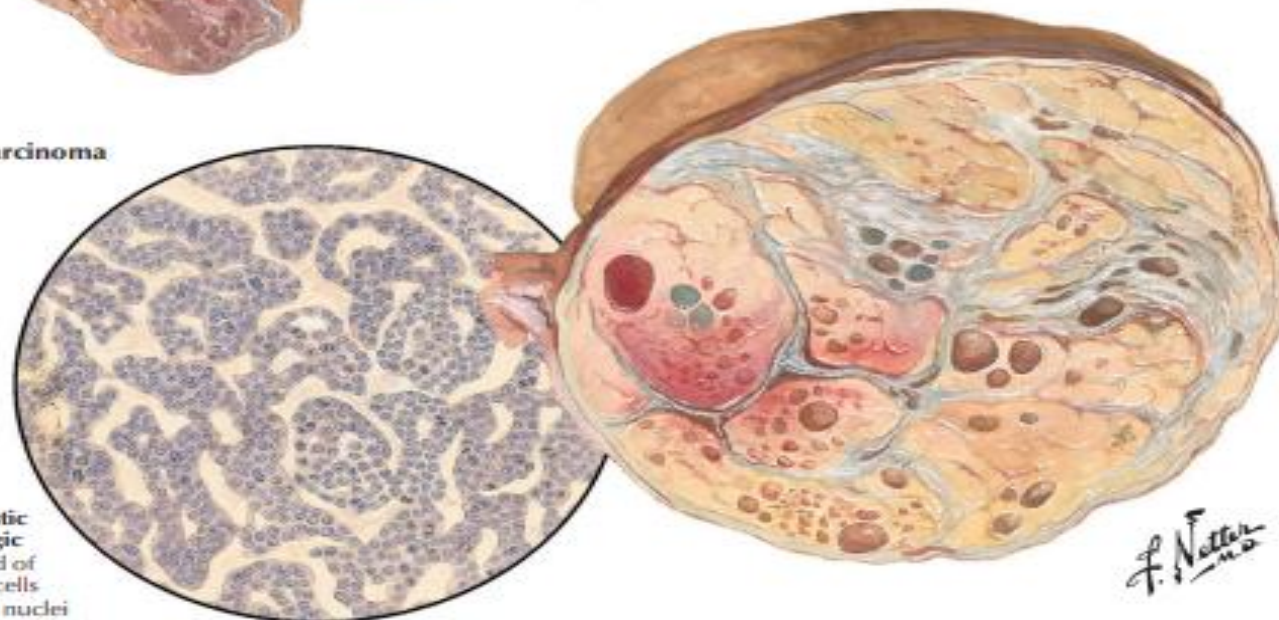


Seminoma



Solid yellow tumor. Composed of uniform cells with single, central nuclei arranged in solid clusters

Embryonal carcinoma



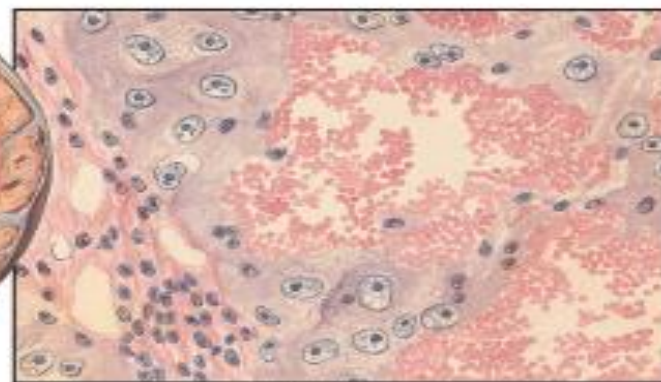
Tumor with cystic and hemorrhagic foci. Composed of large primitive cells with prominent nuclei

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Choriocarcinoma

Cystic
hemorrhagic
component

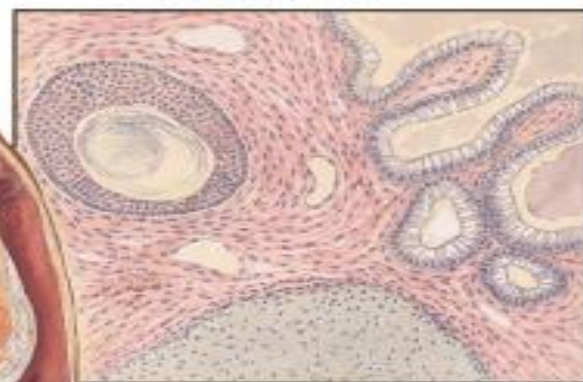
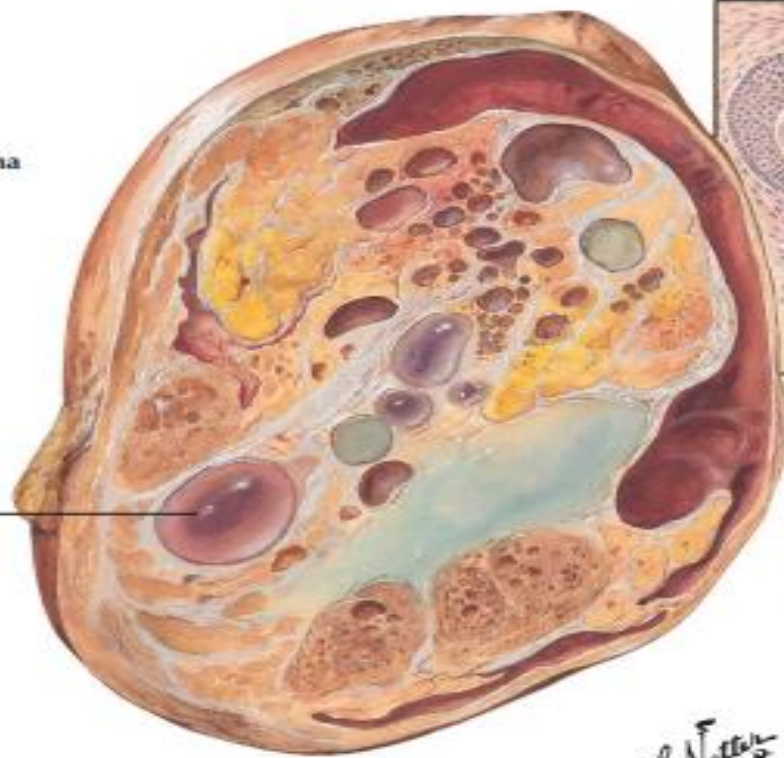
Solid
component



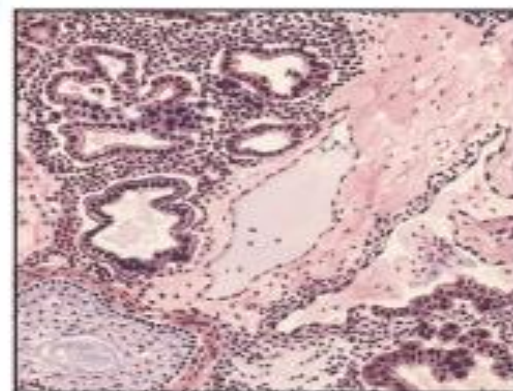
Choriocarcinoma

Teratoma

Glandular
element
containing
inspissated
secretion



Adult teratoma



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NON-SEMINOMATOUS GERM CELL TUMOURS – CONT'D

- **Mixed Germ Cell Tumours** - About 60% of germ cell tumours have more than one of the above histologic types (except spermatocytic seminoma) and are called mixed germ cell tumours.
- clinical behavior worsened by inclusion of more aggressive tumour component in a less malignant tumour.
- Interestingly, metastases of the mixed germ cell tumours may not exactly reproduce the histologic types present in the primary tumour.
- The most common combinations of mixed germ cell tumours are as under:
 - 1. Teratoma, embryonal carcinoma, yolk sac tumour and syncytiotrophoblast.
 - 2. Embryonal carcinoma and teratoma (teratocarcinoma).
 - 3. Seminoma and embryonal carcinoma.

OVERVIEW OF GERM CELL TUMOURS

- Risk factors – cryptorchidism, genetic factors, testicular dysgenesis
- Most present with painless mass of testis
- May present as metastases without a palpable primary mass
- Seminomas remain localized to testis for a long time, metastases occur to lymph nodes
- NSGCTs metastasize earlier, by hematogenous route

OVERVIEW OF GERM CELL TUMOURS

- Seminomas are extremely radiosensitive, NSGCTs are not
- NSGCTs are more aggressive, with a poor prognosis
- NSGCTs first spread to periaortic and iliac LNs, then to mediastinal and left supraclavicular LNs
- ↑incidence of brain mets in choriocarcinoma, and bone metastases in seminoma
- Initial treatment of all testicular germ cell tumours is orchiectomy

OVERVIEW OF GERM CELL TUMOURS

- Chemotherapy to patients with advanced disease
- Classic seminoma treated with radiotherapy + surgery
- Prognosis excellent for classic seminoma, much better for spermatocytic seminoma
- Cure rate 95% in early NSGCTs, 40-90% in met disease
- Therapy & prognosis of the TT are dependant largely on the clinical stage and on the histological type

BIOLOGIC (TUMOUR) MARKERS OF GERM CELL TUMOURS

- Hormones or enzymes produced by germ cell tumours detectable in blood
- Importance of tumour markers
 - Diagnosis of testicular tumours
 - Staging
 - Assessment of tumour burden
 - Monitoring response to treatment

BIOLOGIC (TUMOUR) MARKERS OF GERM CELL TUMOURS – CONT'D

Useful tumour markers in germ cell tumours

- **LDH** - elevation nonspecific. Used to assess tumour burden
- **α -FT** – produced by yolk sac tumours
- **HCG** – choriocarcinoma
- **PLAP** – produced by classic seminoma

TUMORS OF SEX CORD- GONADAL STROMA

INTERSTITIAL (LEYDIG) CELL TUMOR

- Are derived from the **endocrine** cells of the testicular interstitium.
- May be multiple (multicentric) or solitary and unilateral or bilateral.

Gross features

- Neoplasms form circumscribed nodules, usually less than 5 cm in diameter.
- They are rarely large enough to increase the size of the organ.
- They have a distinctive golden brown, homogeneous cut surface.
- The tumor is predisposed to haemorrhage which causes dark discoloration, and to cyst formation in areas.
- The consistency is soft, there being little stroma, and the cut surface is slightly greasy.
- Usually benign

Sertoli cell tumours (Androblastoma)

- These tumors often cause enlargement of the affected testis and there is, sometimes, the development of a feminization syndrome in the patient.
- The feminizing effect of these tumors is due to either their high content of estrogen or to secretion of inhibin.
- Increased **inhibin** secretion by the neoplastic Sertoli cells **inhibits** the secretion of FSH and LH by the pituitary, which, in turn, inhibits testosterone production and presumably alters the ratio of estrogen to testosterone.
- Recovery may follow castration and supportive therapy

Sertoli cell tumours – cont'd

Gross appearance

- Larger tumors are lobulated and enclosed in a tense tunica albuginea
- Small homogenous grey-white to yellow → firm or hard
- The firmness of the tumor is due to the abundance of its stroma (thus different from the other tumors)
- Histologically tumour cells are arranged in a distinctive trabeculae that form cordlike structures & tubules

PROSTATE GLAND

- **PROSTATITIS** – Swelling and inflammation of the prostate gland i.e. prostatitis, may be acute, chronic and granulomatous types
- Causes painful or difficult urination, other symptoms include groin pain, pelvic area pain or genital pain or flu-like symptoms
- Can affect men of all ages but commonly is present in 50 yrs or younger
- **Acute prostatitis** - Acute focal or diffuse suppurative inflammation of the prostate is not uncommon.
- It occurs most commonly due to **ascent of bacteria** from the urethra, less often by **descent from the upper urinary tract** or bladder, and occasionally by lymphogenous or haematogenous spread from a distant focus of infection.

CONT'D

- Common pathogens are those which cause UTI, most frequently **E. coli**, and others such as **Klebsiella**, **Proteus**, **Pseudomonas**, **Enterobacter**, **gonococci**, **staphylococci** and **streptococci**.
- **Chronic bacterial prostatitis** is caused in much the same way and by the same organisms as the acute prostatitis.
- It is generally a **consequence of recurrent UTI**.
- **Chronic abacterial prostatitis** is more common.
- There is no history of recurrent UTI and culture of urine and prostatic secretions is always negative

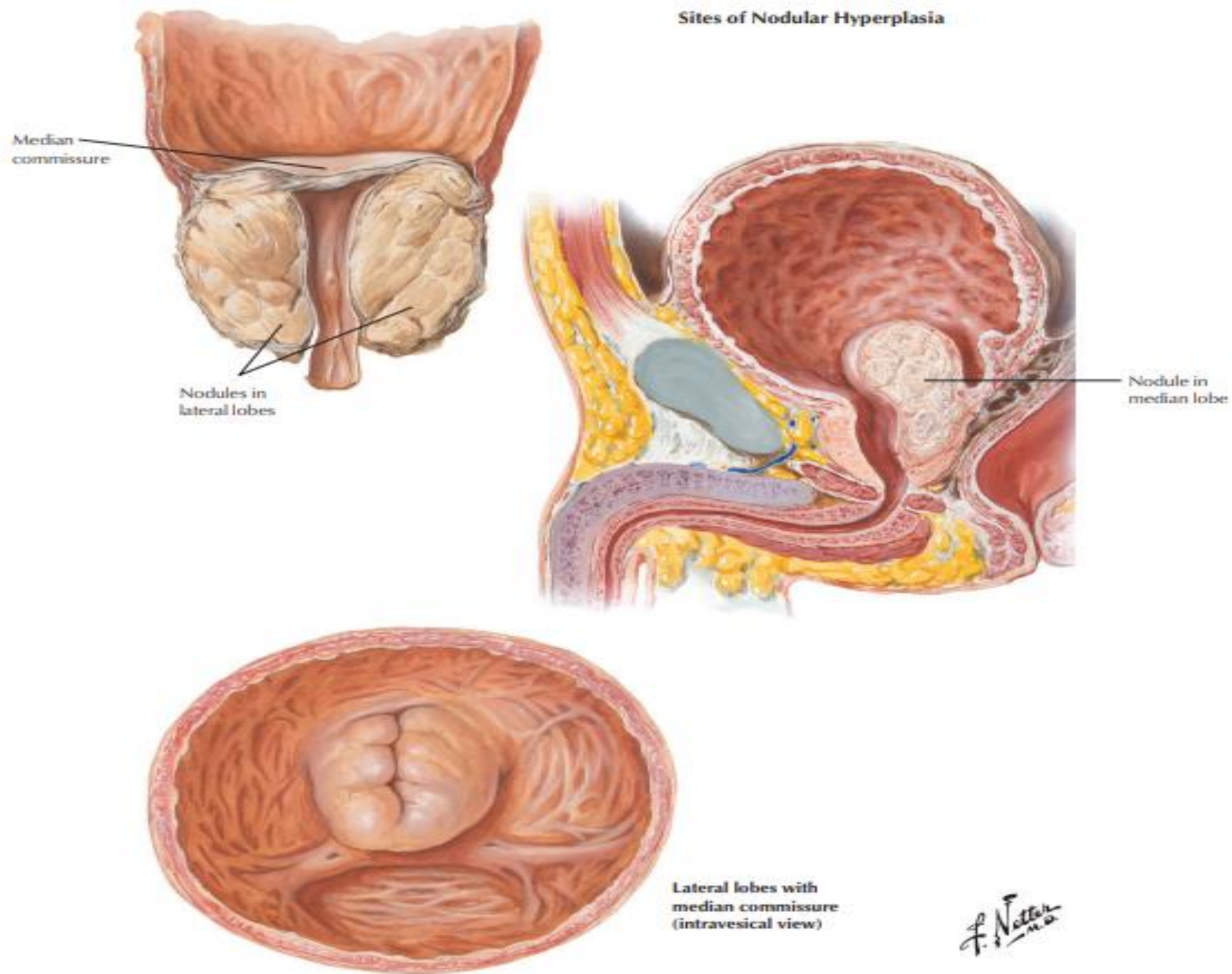
BENIGN PROSTATIC HYPERPLASIA

- Non-neoplastic tumour-like enlargement of the prostate, commonly termed *benign nodular hyperplasia* (BNH) or *benign enlargement of prostate* (BEP)
- A very common condition in men and considered by some as normal ageing process
- Results in ↑ # cells the prostate resulting ↓ diameter of the prostatic urethra
- The cells of the transitional zone under **hyperplasia** increasingly more frequent above the age 50 years
- Cause of BEP has not been fully established but associated few etiologic factors such as **endocrinologic**, **racial**, **inflammation** and **arteriosclerosis** have been implicated

ETIOLOGY & PATHOGENESIS

- The **periurethral inner** prostate which is primarily involved in BEP is responsive to the rising level of **oestrogen**, whereas the outer prostate which is mainly involved in the carcinoma is **responsive to androgen**.
- A plausible hypothesis suggested is that there is synergistic stimulation of the prostate by both hormones—the oestrogen acting to sensitise the prostatic tissue to the growth promoting effect of ***dihydroxy-testosterone*** derived from plasma testosterone.

Sites of Nodular Hyperplasia



Clinical features

- Clinically, the symptomatic cases develop symptoms due to complications such as urethral obstruction and secondary effects on the;
- **Bladder** (e.g. hypertrophy, cystitis),
- **Ureter** (e.g. hydroureter)
- **kidneys** (e.g. hydronephrosis).

The presenting features include

- **frequency**
- **Nocturia**
- **Difficulty in micturition,**
- **Pain**
- **Haematuria**

CARCINOMA OF PROSTATE

- It is a disease of men above the age of 50 years and its prevalence increases with increasing age so that more than 50% of men above 80 yrs are asymptomatic
- Second most common cancer in men after lung cancer
- 1. *Latent carcinoma*.
- This is found unexpectedly as a small focus of carcinoma in the prostate during autopsy studies in men dying of other causes. Its incidence in autopsies has been variously reported as 25-35%.
- 2. *Incidental carcinoma*
- About 15-20% of prostatectomies done for BEP reveal incidental carcinoma of the prostate.

Cont'd

- 3. *Occult carcinoma*
- This is the type in which the patient has **no symptoms of prostatic carcinoma** but shows evidence of metastases on **clinical examination and investigations**.
- 4. *Clinical carcinoma*
- Clinical prostatic carcinoma is the type detected by **rectal examination** and other investigations and confirmed by pathologic examination of **biopsy of the prostate**

MORPHOLOGICAL FEATURES

- Grossly, the prostate may be **enlarged**, **normal in size** or **smaller than normal**
- In 95% of cases, prostatic carcinoma is located in **the peripheral zone**, especially in the posterior lobe.
- The malignant prostate is **firm** and **fibrous**.
- Cut section is **homogeneous** and contains **irregular yellowish areas**.

PROSTATIC ADENOCARCINOMA

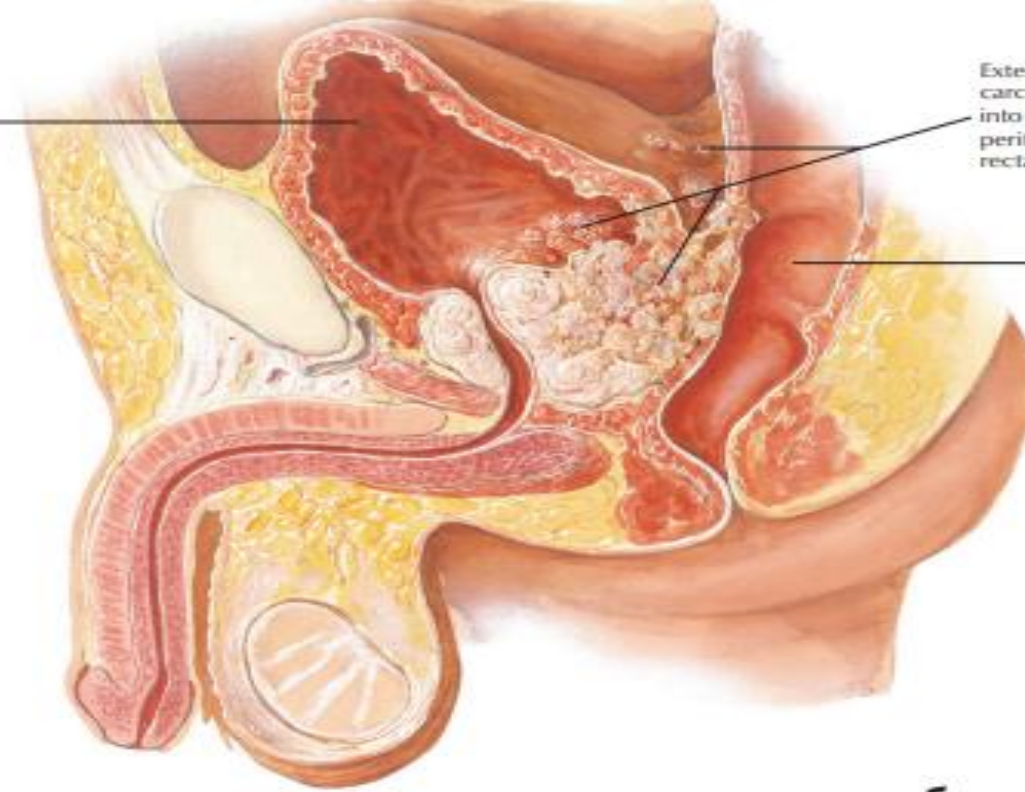


Advanced carcinoma of the prostate

Bladder

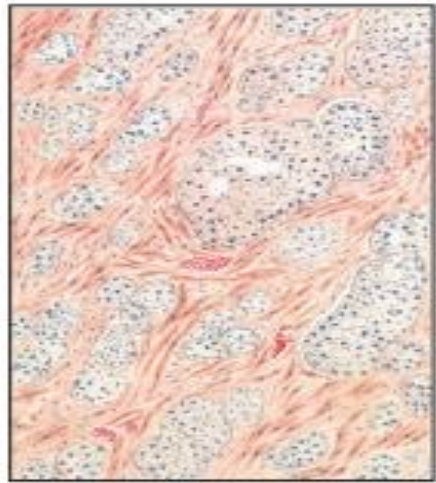
Extension of carcinoma into bladder, peritoneum, and rectal wall

Rectum

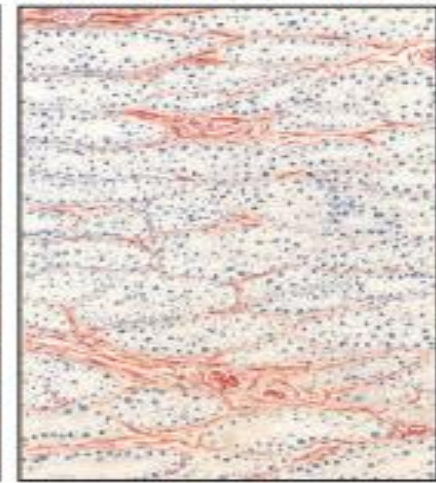


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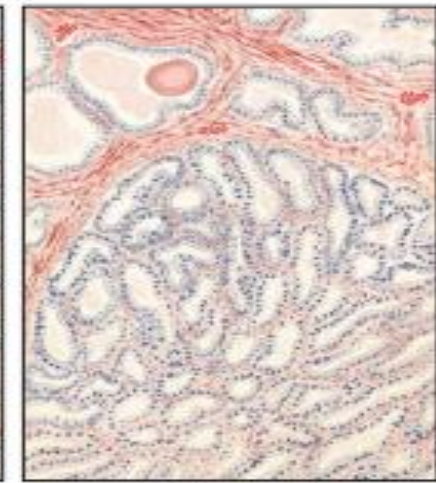
Carcinoma of the prostate: Histology



Scirrhous



Medullary



Carcinoma and benign hyperplasia

CLINICAL FEATURES

- Mostly **has no symptoms because** it affects the peripheral zone of the prostate gland
- In such symptomatic cases, clinical features are: **urinary obstruction with dysuria, frequency, retention of urine, haematuria**, and in 10% of cases pain in the back due to **skeletal metastases**.
- **PSA** is the most **important test** used in the **diagnosis and management of prostate cancer** (4ng/mL)
- Treatment of prostatic carcinoma consists of **surgery, radiotherapy and hormonal therapy**.

THE END!

Questions?